

Fundamental Theorem of Matrix Product States

A Formalization Blueprint

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Chapter 1

Introduction

Chapter 2

Matrix Product States

Fix a physical dimension $d \geq 1$ and a bond dimension $D \geq 1$. All MPS in this blueprint are translation-invariant with periodic boundary conditions (PBC).

2.1 Basic definitions

Definition 2.1 (MPS tensor). A (translation-invariant, PBC) *MPS tensor* with physical dimension d and bond dimension D is a collection of matrices

$$\{A^i\}_{i=0}^{d-1}, \quad A^i \in M_D(\mathbb{C}),$$

indexed by a physical index $i \in \{0, \dots, d-1\}$.

Definition 2.2 (Word evaluation). Given a word $w = (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$, the *word evaluation* is the matrix product

$$A^w := A^{i_1} A^{i_2} \dots A^{i_L} \in M_D(\mathbb{C}),$$

with the convention that the empty word evaluates to the identity: $A^\emptyset = \mathbb{1}_D$.

Lemma 2.3 (Word evaluation respects concatenation). *For any words w_1, w_2 , $A^{w_1 w_2} = A^{w_1} A^{w_2}$.*

Proof. Induction on w_1 . □

Definition 2.4 (Trace coefficient). The *trace coefficient* of a word w is $c_w(A) := \text{tr}(A^w)$.

Definition 2.5 (Matrix product vector). The *matrix product vector* (MPV) of an MPS tensor A at system size N is the vector

$$|V^{(N)}(A)\rangle = \sum_{i_1, \dots, i_N=0}^{d-1} \text{tr}(A^{i_1} A^{i_2} \dots A^{i_N}) |i_1, \dots, i_N\rangle \in (\mathbb{C}^d)^{\otimes N}.$$

The *MPV family* generated by A is the collection $\mathcal{V}(A) = \{|V^{(N)}(A)\rangle\}_{N \geq 1}$.

Definition 2.6 (Transfer map). The *transfer map* associated to an MPS tensor A is the linear map $\mathcal{E}_A : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$\mathcal{E}_A(X) = \sum_{i=0}^{d-1} A^i X (A^i)^\dagger.$$

Lemma 2.7 (Transfer map is positive). *If $X \geq 0$, then $\mathcal{E}_A(X) \geq 0$.*

Proof. Each summand $A^i X (A^i)^\dagger$ is PSD when X is PSD. $\square$

2.2 Gauge equivalence and same MPV

Definition 2.8 (Gauge equivalence). Two MPS tensors A and B of the same bond dimension D are *gauge equivalent* if there exists an invertible matrix $X \in \text{GL}_D(\mathbb{C})$ such that

$$B^i = X A^i X^{-1} \quad \text{for all } i \in \{0, \dots, d-1\}.$$

Definition 2.9 (Same MPV). Two MPS tensors A and B (of the same bond dimension) *generate the same MPV family* if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every system size $N \geq 1$.

Definition 2.10 (Same MPV — different bond dimensions). For MPS tensors A and B of possibly different bond dimensions D_1 and D_2 , we write $\mathcal{V}(A) = \mathcal{V}(B)$ if $|V^{(N)}(A)\rangle = |V^{(N)}(B)\rangle$ for every $N \geq 1$.

Definition 2.11 (Proportional MPV). Two MPS tensors A and B (of possibly different bond dimensions) *generate proportional MPV families* if for every N there exists $c_N \in \mathbb{C}$ such that $|V^{(N)}(A)\rangle = c_N |V^{(N)}(B)\rangle$.

Lemma 2.12 (Word evaluation under scaling). *Let ζA denote the tensor with matrices $\{\zeta A^i\}_{i=0}^{d-1}$. Then for any word w ,*

$$(\zeta A)^w = \zeta^{|w|} A^w,$$

where $|w|$ is the length of w .

Proof. Induction on w : the base case gives $\zeta^0 = 1$, and each step picks up one factor of ζ . $\square$

Definition 2.13 (Gauge-phase equivalence). Two MPS tensors A and B are *gauge-phase equivalent* if there exist $X \in \text{GL}_D(\mathbb{C})$ and $\zeta \in \mathbb{C} \setminus \{0\}$ such that

$$B^i = \zeta X A^i X^{-1} \quad \text{for all } i.$$

The requirement $\zeta \neq 0$ ensures that gauge-phase equivalence preserves injectivity.

Lemma 2.14 (Word evaluation under conjugation). *If $B^i = X A^i X^{-1}$ for all i , then for every word w ,*

$$B^w = X A^w X^{-1}.$$

Proof. Induction on the word length, using $XX^{-1} = \mathbb{1}$. $\square$

Theorem 2.15 (Gauge invariance of MPVs). *If A and B are gauge equivalent, then they generate the same MPV family.*

Proof. Let $B^i = X A^i X^{-1}$. By Lemma 2.14, $B^w = X A^w X^{-1}$ for every word w . Then $\text{tr}(B^w) = \text{tr}(X A^w X^{-1}) = \text{tr}(A^w)$ by cyclicity of the trace. $\square$

2.3 Injectivity and normality

Definition 2.16 (Injectivity). An MPS tensor A is *injective* if its matrices span the full matrix algebra: $\text{span}_{\mathbb{C}}\{A^i : i = 0, \dots, d-1\} = M_D(\mathbb{C})$.

Definition 2.17 (L -block injectivity). An MPS tensor A is *L -block injective* if $\text{span}_{\mathbb{C}}\{A^{i_1} \dots A^{i_L} : (i_1, \dots, i_L) \in \{0, \dots, d-1\}^L\} = M_D(\mathbb{C})$. An injective tensor is 1-block injective.

Definition 2.18 (Normal tensor). An MPS tensor A is *normal* [CPGSV21] if there exists a finite L_0 such that A is L_0 -block injective.

2.4 Canonical form

Definition 2.19 (Canonical form). A *canonical form* for an MPS consists of:

1. a number of blocks $r \geq 1$,
2. bond dimensions $D_1, \dots, D_r$,
3. injective tensors $\{A_k^i\}_{i=0}^{d-1}$ with $A_k^i \in M_{D_k}(\mathbb{C})$ for each block $k \in \{1, \dots, r\}$,
4. scaling factors $\mu_1, \dots, \mu_r \in \mathbb{C}$.

The associated MPS tensor is the block-diagonal tensor

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

This decomposition is defined in [PGVWC07] and reviewed in [CPGSV21].

Remark 2.20. The formalization requires each block tensor to be injective (Definition 2.16). More generally, canonical forms in the literature [CPGSV21] allow normal tensors (Definition 2.18), which become injective after sufficient blocking (Definition 2.23).

Definition 2.21 (Block-diagonal MPS tensor). Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, the *block-diagonal MPS tensor* is the tensor of total bond dimension $D = \sum_{k=1}^r D_k$ defined by

$$A^i = \bigoplus_{k=1}^r \mu_k A_k^i.$$

Theorem 2.22 (MPV decomposition). *The MPV of a block-diagonal tensor decomposes as*

$$|V^{(N)}(A)\rangle = \sum_{k=1}^r \mu_k^N |V^{(N)}(A_k)\rangle.$$

Proof. For each physical index i , the block-diagonal tensor $A^i = \bigoplus_k \mu_k A_k^i$ is a block-diagonal matrix. By Lemma 2.12, the word evaluation of each block picks up a factor $\mu_k^{|w|}$, and the full word evaluation is block-diagonal with blocks $\mu_k^N A_k^w$. Taking the trace of a block-diagonal matrix gives the sum of the block traces: $\text{tr}(A^w) = \sum_k \mu_k^N \text{tr}(A_k^w)$. $\square$

2.5 Blocking

Definition 2.23 (Blocked tensor). Given an MPS tensor A and a blocking length $L \geq 1$, the L -blocked tensor is the MPS tensor with physical dimension d^L and the same bond dimension D , whose matrices are indexed by words $(i_1, \dots, i_L) \in \{0, \dots, d-1\}^L$:

$$(A^{[L]})^{(i_1, \dots, i_L)} = A^{i_1} A^{i_2} \dots A^{i_L}.$$

Lemma 2.24 (Blocked word evaluation). Let $w = (J_1, \dots, J_m)$ be a word in the blocked alphabet (each $J_k \in \{0, \dots, d^L-1\}$). Decoding each blocked index J_k back to an L -tuple of physical indices and concatenating gives a word $\tilde{w}$ of length mL in the original alphabet. Then $(A^{[L]})^w = A^{\tilde{w}}$.

Proof. Induction on w , using the concatenation lemma (Lemma 2.3). $\square$

Lemma 2.25 (Blocking preserves MPV coefficients). For each basis state $\sigma = (\sigma_1, \dots, \sigma_N)$ of the blocked system (with $\sigma_k \in \{0, \dots, d^L-1\}$), there exists a basis state $\tilde{\sigma} = (\tilde{\sigma}_1, \dots, \tilde{\sigma}_{NL})$ of the original system such that

$$V^{(N)}(A^{[L]})_\sigma = V^{(NL)}(A)_{\tilde{\sigma}}.$$

Proof. Flatten the blocked basis state and apply the blocked word evaluation lemma (Lemma 2.24). $\square$

Theorem 2.26 (Blocking preserves same MPV). If A and B generate the same MPV family, then for any blocking length L , the blocked tensors $A^{[L]}$ and $B^{[L]}$ also generate the same MPV family.

Proof. By Lemma 2.25, each MPV coefficient of the blocked tensor $A^{[L]}$ equals a coefficient of A at the flattened configuration. The same holds for $B^{[L]}$ (with the same flattened configuration). Since A and B agree on all configurations, so do $A^{[L]}$ and $B^{[L]}$. $\square$

2.6 MPV overlap

Definition 2.27 (MPV as Hilbert-space vector). We regard $|V^{(N)}(A)\rangle$ as an element of $\mathbb{C}^{d^N}$ indexed by configurations $\sigma \in \{0, \dots, d-1\}^N$, with σ -component $V^{(N)}(A)_\sigma$ as in Definition 2.5. This allows us to use the standard Euclidean inner product on $\mathbb{C}^{d^N}$.

Definition 2.28 (MPV inner product). The MPV inner product between two MPS tensors A and B at system size N is

$$\langle V^{(N)}(A) | V^{(N)}(B) \rangle := \sum_{\sigma \in \{0, \dots, d-1\}^N} \overline{V^{(N)}(A)_\sigma} V^{(N)}(B)_\sigma,$$

i.e. the Euclidean inner product on $\mathbb{C}^{d^N}$ (conjugate-linear in the first argument).

Definition 2.29 (MPV overlap). The MPV overlap between two MPS tensors A (of bond dimension D_1) and B (of bond dimension D_2) at system size N is

$$O_{AB}(N) := \sum_{\sigma \in \{0, \dots, d-1\}^N} V^{(N)}(A)_\sigma \overline{V^{(N)}(B)_\sigma}.$$

Lemma 2.30 (Overlap equals conjugate inner product). $O_{AB}(N) = \overline{\langle V^{(N)}(A) | V^{(N)}(B) \rangle}$.

Proof. The overlap sums $V_\sigma \overline{W_\sigma}$ while the inner product sums $\overline{V_\sigma} W_\sigma$. Complex conjugation swaps these. $\square$

Chapter 3

The Single-Block Fundamental Theorem

This chapter proves that if an injective MPS tensor A generates the same MPV family as another tensor B (of the same bond dimension), then A and B are gauge equivalent.

3.1 Trace pairing

Theorem 3.1 (Nondegeneracy of the trace pairing). *For $M \in M_D(\mathbb{C})$, $\text{tr}(MN) = 0$ for all $N \in M_D(\mathbb{C})$ if and only if $M = 0$.*

Proof. Testing against the matrix units E_{ji} (the matrix with 1 in position (j, i) and 0 elsewhere) gives $\text{tr}(E_{ji}M) = M_{ij}$. Hence if $\text{tr}(MN) = 0$ for all N , then every entry of M vanishes. $\square$

Lemma 3.2 (Same MPV implies trace agreement). *If A and B generate the same MPV family, then $\text{tr}(A^w) = \text{tr}(B^w)$ for every word w .*

Proof. Viewing the word w as a configuration of length $|w|$, the MPV coefficient at this configuration is exactly $\text{tr}(A^w)$. The same-MPV hypothesis gives equality at every system size. $\square$

Lemma 3.3 (Length-2 trace identity). *If A and B generate the same MPV family, then $\text{tr}(A^i A^j) = \text{tr}(B^i B^j)$ for all $i, j \in \{0, \dots, d-1\}$.*

Proof. Apply Lemma 3.2 to the word $w = (i, j)$. $\square$

Definition 3.4 (Trace pairing map). For an MPS tensor A , define the linear map $\Phi_A : M_D(\mathbb{C}) \rightarrow \mathbb{C}^d$ by

$$\Phi_A(M)_i = \text{tr}(M \cdot A^i).$$

Theorem 3.5 (Injectivity of the trace pairing map). *If A is injective, then $\ker \Phi_A = \{0\}$.*

Proof. If $\Phi_A(M) = 0$, then $\text{tr}(M \cdot A^i) = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, this gives $\text{tr}(MN) = 0$ for all N , so $M = 0$ by Theorem 3.1. $\square$

Lemma 3.6 (Injectivity transfer via range inclusion). *If $\Phi_A, \Phi_B : V \rightarrow W$ are linear maps with $\ker \Phi_A = \{0\}$ and $\text{range}(\Phi_A) \subseteq \text{range}(\Phi_B)$, then $\ker \Phi_B = \{0\}$.*

Proof. By rank-nullity, $\ker \Phi_A = \{0\}$ gives $\dim(\text{range}(\Phi_A)) = \dim V$. The range inclusion forces $\dim(\text{range}(\Phi_B)) \geq \dim V$, so $\dim(\ker \Phi_B) = 0$ by rank-nullity again. $\square$

Theorem 3.7 (Nonvanishing trace on injective tensors). *If $D \geq 1$, A is injective, A and B generate the same MPV family, and $B^i = 0$ for all i , then we have a contradiction.*

Proof. If $B^i = 0$ for all i , then Lemma 3.2 gives $\text{tr}(A^w) = 0$ for every word w , in particular for all length-1 words. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, the trace functional vanishes identically, contradicting $\text{tr}(\mathbb{1}_D) = D \neq 0$. $\square$

3.2 Linear extension

Theorem 3.8 (Existence and uniqueness of the linear extension). *If A is injective and A, B generate the same MPV family, then there exists a unique linear map $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ satisfying $T(A^i) = B^i$ for all i .*

Proof. Write $\ell_A : \mathbb{C}^d \rightarrow M_D(\mathbb{C})$ for the map that sends a coefficient vector c to $\sum_i c_i A^i$ (linear combination with the A^i). By the length-2 trace identity (Lemma 3.3), $\Phi_A \circ \ell_A = \Phi_B \circ \ell_B$. Since A is injective, ℓ_A is surjective, giving $\text{range}(\Phi_A) \subseteq \text{range}(\Phi_B)$. By Theorem 3.5, $\ker \Phi_A = \{0\}$; Lemma 3.6 then forces $\ker \Phi_B = \{0\}$. A left inverse g of Φ_B gives $T := g \circ \Phi_A$, and uniqueness follows because the $\{A^i\}$ span $M_D(\mathbb{C})$. $\square$

Theorem 3.9 (Multiplicativity of the linear extension). *The linear extension T from Theorem 3.8 is multiplicative: $T(MN) = T(M)T(N)$ for all $M, N \in M_D(\mathbb{C})$.*

Proof. **Step 1** ($\Phi_B \circ T = \Phi_A$). By the length-2 trace identity (Lemma 3.3), $\Phi_B(T(A^i)) = \Phi_B(B^i) = \Phi_A(A^i)$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\Phi_B \circ T = \Phi_A$.

Step 2 ($T(A^i A^j) = B^i B^j$). The length-3 trace identity gives $\text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k)$ for all i, j, k . So $\Phi_B(T(A^i A^j))_k = \Phi_A(A^i A^j)_k = \text{tr}(A^i A^j A^k) = \text{tr}(B^i B^j B^k) = \Phi_B(B^i B^j)_k$. Injectivity of Φ_B (from Lemma 3.6) gives $T(A^i A^j) = B^i B^j$.

Step 3 (bilinear extension). For fixed j , the maps $M \mapsto T(M \cdot A^j)$ and $M \mapsto T(M) \cdot B^j$ agree on the spanning set $\{A^i\}$, hence on all of $M_D(\mathbb{C})$. Applying this once more with j replaced by a general matrix gives full multiplicativity. $\square$

3.3 Simplicity and Skolem–Noether

Theorem 3.10 (Simplicity of $M_D(\mathbb{C})$). *A nonzero multiplicative $\mathbb{C}$ -linear endomorphism of $M_D(\mathbb{C})$ is bijective.*

Proof. The kernel of a multiplicative linear map is a two-sided ideal. Since $M_D(\mathbb{C})$ is a simple ring (it has no nontrivial two-sided ideals), the kernel is either $\{0\}$ or $M_D(\mathbb{C})$. The latter would force the map to be zero, contradicting the hypothesis. Hence the map is injective, and surjectivity follows from finite-dimensionality. $\square$

Theorem 3.11 (Skolem–Noether). *Every $\mathbb{C}$ -algebra automorphism of $M_D(\mathbb{C})$ is inner: for any $f \in \text{Aut}_{\mathbb{C}\text{-alg}}(M_D(\mathbb{C}))$, there exists $X \in \text{GL}_D(\mathbb{C})$ such that $f(M) = XMX^{-1}$ for all M .*

Proof. Via the canonical identification $M_D(\mathbb{C}) \cong \text{End}_{\mathbb{C}}(\mathbb{C}^D)$, the automorphism f induces an automorphism of $\text{End}_{\mathbb{C}}(\mathbb{C}^D)$. Every algebra automorphism of $\text{End}(V)$ is conjugation by an invertible linear map (this is a standard result in the theory of central simple algebras). Translating back to matrices gives the invertible X . $\square$

Lemma 3.12 (Promotion to algebra homomorphism). *If $T : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a multiplicative surjective $\mathbb{C}$ -linear map, then $T(\mathbb{1}) = \mathbb{1}$, and T is a $\mathbb{C}$ -algebra homomorphism.*

Proof. By surjectivity, there exists X with $T(X) = \mathbb{1}$. Then $T(\mathbb{1}) = T(X) \cdot T(\mathbb{1}) = T(X \cdot \mathbb{1}) = T(X) = \mathbb{1}$, using multiplicativity and $T(X) = \mathbb{1}$. With $T(\mathbb{1}) = \mathbb{1}$ established, T preserves the unit and hence is a $\mathbb{C}$ -algebra homomorphism. $\square$

3.4 Assembly

Theorem 3.13 (Single-block Fundamental Theorem). *Let A be an injective MPS tensor and let B be an MPS tensor of the same bond dimension. If A and B generate the same MPV family, then they are gauge equivalent: there exists $X \in \text{GL}_D(\mathbb{C})$ such that $B^i = XA^iX^{-1}$ for all i .*

Proof. By Theorem 3.8, there is a unique linear T with $T(A^i) = B^i$. By Theorem 3.9, T is multiplicative. T is nonzero: if $T = 0$ then $B^i = 0$ for all i , contradicting Theorem 3.7. By Theorem 3.10, T is bijective. By Lemma 3.12, T is a $\mathbb{C}$ -algebra automorphism. Theorem 3.11 gives an invertible X with $T(M) = XMX^{-1}$, and evaluating at $M = A^i$ yields $B^i = XA^iX^{-1}$. $\square$

Chapter 4

Quantum Channels and Positive Maps

This chapter develops the theory of positive maps and quantum channels on matrix algebras, following [Wol12].

4.1 Positive maps and channels

Definition 4.1 (Positive map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *positive* if $E(X) \geq 0$ whenever $X \geq 0$.

Definition 4.2 (Trace-preserving map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *trace-preserving* if $\text{tr}(E(X)) = \text{tr}(X)$ for all X .

Definition 4.3 (Completely positive map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *completely positive* (CP) if it admits a *Kraus representation*: there exist operators $\{K_i\}_{i=0}^{r-1}$ with $K_i \in M_D(\mathbb{C})$ such that

$$E(X) = \sum_{i=0}^{r-1} K_i X K_i^\dagger \quad \text{for all } X \in M_D(\mathbb{C}).$$

By Choi's theorem, this is equivalent to $E \otimes \text{id}_n$ being positive for all n .

Theorem 4.4 (CP maps are positive). *Every completely positive map is positive.*

Proof. If $X \geq 0$, then each summand $K_i X K_i^\dagger \geq 0$ (conjugation preserves positive semidefiniteness), so the sum is PSD. $\square$

Definition 4.5 (Quantum channel). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is a *quantum channel* if it is both completely positive and trace-preserving (CPTP), following [Wol12].

Definition 4.6 (Density matrices). The set of *density matrices* in $M_D(\mathbb{C})$ is

$$\mathcal{D}_D = \{\rho \in M_D(\mathbb{C}) : \rho \geq 0, \text{tr}(\rho) = 1\}.$$

Theorem 4.7 (Density matrices are compact). $\mathcal{D}_D$ is compact (in the entry-wise topology on $M_D(\mathbb{C})$).

Proof. The PSD cone is closed (preimage of the closed nonneg cone under continuous quadratic forms), and PSD matrices with $\text{tr}(\rho) = 1$ have entry norms bounded by 1 (diagonal entries by trace; off-diagonal by Cauchy–Schwarz). Heine–Borel gives compactness. $\square$

Theorem 4.8 (Density matrices are convex). $\mathcal{D}_D$ is convex.

Proof. Convex combination of PSD matrices is PSD, and trace is linear. $\square$

Theorem 4.9 (Density matrices are nonempty). For $D \geq 1$, $\mathcal{D}_D \neq \emptyset$.

Proof. $\frac{1}{D}\mathbb{1}_D$ is a density matrix. $\square$

Theorem 4.10 (Channels preserve density matrices). If E is a quantum channel, then $E(\mathcal{D}_D) \subseteq \mathcal{D}_D$.

Proof. Complete positivity implies positivity (Theorem 4.4), so $E(\rho) \geq 0$; trace preservation gives $\text{tr}(E(\rho)) = \text{tr}(\rho) = 1$. $\square$

4.2 Kadison–Schwarz inequality

A Kraus map is the concrete realisation of a completely positive map (Definition 4.3). The Kadison–Schwarz inequality and the multiplicative domain characterisation are proved for Kraus maps, then applied to transfer maps, which are Kraus maps by construction (Theorem 4.24).

Definition 4.11 (Kraus map). Given operators $\{K_i\}_{i=0}^{d-1}$ with $K_i \in M_D(\mathbb{C})$, the *Kraus map* is $E(X) = \sum_{i=0}^{d-1} K_i X K_i^\dagger$. A linear map is completely positive (Definition 4.3) if and only if it can be written in this form.

Definition 4.12 (Adjoint Kraus map). The *adjoint Kraus map* is $E^*(X) = \sum_{i=0}^{d-1} K_i^\dagger X K_i$. When the $\{K_i\}$ are the matrices of an MPS tensor A , the adjoint Kraus map corresponds to the transfer map $\mathcal{E}_A$ with the roles of A^i and $(A^i)^\dagger$ exchanged.

Definition 4.13 (Unital Kraus map). A Kraus map is *unital* if $\sum_{i=0}^{d-1} K_i K_i^\dagger = \mathbb{1}$.

Definition 4.14 (Trace-preserving Kraus map). A Kraus map is *trace-preserving* if $\sum_{i=0}^{d-1} K_i^\dagger K_i = \mathbb{1}$. This is the standard MPS normalization condition.

Theorem 4.15 (Kadison–Schwarz inequality). Let $E(X) = \sum_i K_i X K_i^\dagger$ be a unital Kraus map (i.e. $\sum_i K_i K_i^\dagger = \mathbb{1}$). Then for all $X \in M_D(\mathbb{C})$,

$$E(X^\dagger X) \geq E(X)^\dagger E(X).$$

This is [Wol12, Eq. (5.2)].

Proof. The 2×2 block matrix $\begin{bmatrix} X^\dagger X & X^\dagger \\ X & \mathbb{1} \end{bmatrix}$ is PSD ($= vv^\dagger$ with $v = [X^\dagger, \mathbb{1}]^T$). Applying the unital CP map blockwise preserves PSD-ness (each block $K_i(\cdot)K_i^\dagger$ preserves PSD, and unitality ensures the $(2, 2)$ -block maps to $\mathbb{1}$). The Schur complement of the $(2, 2)$ -block $\mathbb{1}$ gives the result. $\square$

Theorem 4.16 (Kadison–Schwarz for the adjoint channel). If $\sum_i K_i^\dagger K_i = \mathbb{1}$ (trace-preserving), then the adjoint Kraus map satisfies $E^*(X^\dagger X) \geq E^*(X)^\dagger E^*(X)$.

Proof. The adjoint operators $\{K_i^\dagger\}$ satisfy $\sum_i K_i^\dagger (K_i^\dagger)^\dagger = \sum_i K_i^\dagger K_i = \mathbb{1}$, so they form a unital family. Apply Theorem 4.15 to $\{K_i^\dagger\}$. $\square$

Theorem 4.17 (Hilbert–Schmidt contraction). *If a Kraus map is both unital and trace-preserving, then $\text{tr}(E(X)^\dagger E(X)) \leq \text{tr}(X^\dagger X)$ for all $X \in M_D(\mathbb{C})$.*

Proof. By Theorem 4.15, $E(X^\dagger X) - E(X)^\dagger E(X) \geq 0$, so $\text{tr}(E(X^\dagger X)) \geq \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. $\square$

4.3 Multiplicative domain

Theorem 4.18 (KS gap decomposition). *For a unital Kraus map E , the Kadison–Schwarz gap decomposes as*

$$E(X^\dagger X) - E(X)^\dagger E(X) = \sum_{i=0}^{d-1} R_i^\dagger R_i,$$

where $R_i := XK_i^\dagger - K_i^\dagger E(X)$.

Proof. Expand the right-hand side $\sum_i (XK_i^\dagger - K_i^\dagger E(X))^\dagger (XK_i^\dagger - K_i^\dagger E(X))$, distribute, and use unitality $\sum_i K_i K_i^\dagger = \mathbb{1}$ to identify the result with the KS gap. $\square$

Theorem 4.19 (KS equality implies Kraus commutation). *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$ for some X , then $XK_i^\dagger = K_i^\dagger E(X)$ for all i .*

Proof. By Theorem 4.18, the KS gap is $\sum_i R_i^\dagger R_i$ with $R_i = XK_i^\dagger - K_i^\dagger E(X)$. If the gap vanishes, $\sum_i R_i^\dagger R_i = 0$. Each summand $R_i^\dagger R_i$ is PSD, so each must be zero, giving $R_i = 0$. $\square$

Theorem 4.20 (KS equality for peripheral eigenvectors). *Let E be a Kraus map that is both unital and trace-preserving. If $E(X) = \mu X$ with $|\mu| = 1$, then the KS gap vanishes: $E(X^\dagger X) = E(X)^\dagger E(X)$.*

Proof. The gap $G := E(X^\dagger X) - E(X)^\dagger E(X)$ is PSD by Theorem 4.15. Its trace satisfies $\text{tr}(G) = \text{tr}(E(X^\dagger X)) - \text{tr}(E(X)^\dagger E(X))$. Trace preservation gives $\text{tr}(E(X^\dagger X)) = \text{tr}(X^\dagger X)$. Using $E(X) = \mu X$ and $|\mu| = 1$, $\text{tr}(E(X)^\dagger E(X)) = |\mu|^2 \text{tr}(X^\dagger X) = \text{tr}(X^\dagger X)$. So $\text{tr}(G) = 0$, and a PSD matrix with zero trace must be zero. $\square$

Theorem 4.21 (Multiplicative domain). *Let E be a unital Kraus map. If $E(X^\dagger X) = E(X)^\dagger E(X)$, then $E(X^\dagger Y) = E(X)^\dagger E(Y)$ for all $Y \in M_D(\mathbb{C})$.*

Proof. From Theorem 4.19, $XK_i^\dagger = K_i^\dagger E(X)$ for all i . Taking conjugate transposes: $K_i X^\dagger = E(X)^\dagger K_i$. Then

$$\begin{aligned} E(X^\dagger Y) &= \sum_i K_i X^\dagger Y K_i^\dagger \\ &= \sum_i E(X)^\dagger K_i Y K_i^\dagger \\ &= E(X)^\dagger E(Y). \square \end{aligned}$$

4.4 Irreducibility

Definition 4.22 (Irreducible map). A linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ is *irreducible* if whenever P is an orthogonal projection satisfying $E(PM_D(\mathbb{C})P) \subseteq PM_D(\mathbb{C})P$, then $P = 0$ or $P = \mathbb{1}$. This is [Wol12, Thm. 6.6, condition (1)]. The definition applies to any linear map; complete positivity is not required.

Theorem 4.23 (Injectivity implies irreducibility). *If A is an injective MPS tensor, then its transfer map $\mathcal{E}_A$ (which is CP by Theorem 4.24) is irreducible.*

Proof. If P is an invariant projection for $\mathcal{E}_A$, then $(\mathbb{1} - P)A^iP = 0$ for all i . Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $(\mathbb{1} - P)MP = 0$ for all M , so $P = 0$ or $P = \mathbb{1}$. $\square$

Theorem 4.24 (Transfer map is CP). *The transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is completely positive (its Kraus operators are the matrices $\{A^i\}$).*

Proof. Immediate from the definition: the transfer map is defined as a Kraus map with operators $K_i = A^i$. $\square$

Theorem 4.25 (Transfer map is a channel). *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, then the transfer map $\mathcal{E}_A(X) = \sum_i A^i X (A^i)^\dagger$ is a quantum channel (CPTP).*

Proof. Complete positivity is Theorem 4.24. Trace preservation: $\text{tr}(\mathcal{E}_A(X)) = \sum_i \text{tr}(A^i X (A^i)^\dagger) = \text{tr}((\sum_i (A^i)^\dagger A^i)X) = \text{tr}(X)$ by cyclicity and the normalization hypothesis. $\square$

4.5 Cesàro fixed-point theorem

Definition 4.26 (Cesàro mean). The *Cesàro mean* of a linear map E at order N , applied to a starting matrix ρ_0 , is

$$\sigma_N = \frac{1}{N} \sum_{n=0}^{N-1} E^n(\rho_0).$$

Theorem 4.27 (Cesàro telescope). $E(\sigma_N) - \sigma_N = \frac{1}{N}(E^N(\rho_0) - \rho_0)$.

Proof. Telescoping: $E(\sigma_N) = \frac{1}{N} \sum_{n=1}^N E^n(\rho_0)$ and subtracting σ_N cancels all but the first and last terms. $\square$

Theorem 4.28 (Hermitian fixed-point decomposition). *Let E be a quantum channel and $X = E(X)$ a Hermitian fixed point with decomposition $X = P_+ - P_-$ into orthogonal positive and negative parts ($P_\pm \geq 0$, $\text{tr}(P_+ P_-) = 0$). Then $E(P_+) = P_+$ and $E(P_-) = P_-$. This is [Wol12, Prop. 6.8].*

Proof. Let Q be the support projection of P_+ . Trace preservation and positivity give $\text{tr}(QE(P_-)) = 0$ and $E(P_-) \geq 0$, so $QE(P_-) = 0$. The fixed-point equation $E(P_+) - E(P_-) = P_+ - P_-$ then identifies $E(P_\pm) = P_\pm$. $\square$

Theorem 4.29 (Cesàro fixed-point theorem). *Every quantum channel $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ (with $D \geq 1$) has a nonzero positive semidefinite fixed point: there exists $\rho \geq 0$, $\rho \neq 0$, with $E(\rho) = \rho$.*

Proof. Start with any $\rho_0 \in \mathcal{D}_D$ (Theorem 4.9). Each Cesàro mean σ_N lies in $\mathcal{D}_D$ (by convexity and channel preservation). By compactness (Theorem 4.7), extract a convergent subsequence $\sigma_{N_k} \rightarrow \rho$. The telescope identity (Theorem 4.27) gives $\|E(\sigma_N) - \sigma_N\| = \frac{1}{N}\|E^N(\rho_0) - \rho_0\| \rightarrow 0$, so $E(\rho) = \rho$ by continuity. In [Wol12, Thm. 6.13], existence is proved via Brouwer's fixed-point theorem; we use the Cesàro argument instead, as Brouwer's theorem is not available in Mathlib. $\square$

4.6 Fixed-point projection

Definition 4.30 (Fixed-point projection). Given a linear map $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ and a fixed point ρ with $E(\rho) = \rho$ and $\text{tr}(\rho) \neq 0$, the *fixed-point projection* is the rank-one map

$$P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)} \rho.$$

This projects onto the span of ρ along the kernel of the trace functional.

Theorem 4.31 (Power decomposition). *If E is trace-preserving and $E(\rho) = \rho$, then for $n \geq 1$,*

$$E^n = P + (E - P)^n,$$

where P is the fixed-point projection.

Proof. P is idempotent ($P^2 = P$), and trace preservation gives $E \circ P = P$ and $P \circ E = P$. These imply $(E - P) \circ P = 0$ and $P \circ (E - P) = 0$, so in the binomial expansion of $(P + (E - P))^n$ all cross terms vanish, leaving $E^n = P^n + (E - P)^n = P + (E - P)^n$. $\square$

4.7 Peripheral spectrum and primitivity

Definition 4.32 (Peripheral spectrum). The *peripheral spectrum* of a bounded linear operator T is the set of spectral values μ with $\|\mu\|$ equal to the spectral radius of T .

Definition 4.33 (Peripheral eigenvalues). The *peripheral eigenvalues* of a linear map E on a finite-dimensional space are the eigenvalues λ on the unit circle: $|\lambda| = 1$. For a channel, the spectral radius is 1 [Wol12, Prop. 6.1], so these coincide with the eigenvalues of maximal modulus.

Note: The Lean definition hardcodes $|\lambda| = 1$ rather than $|\lambda| = \rho(E)$ (the spectral radius). This is correct for channels but would need generalisation for maps with spectral radius $\neq 1$.

Theorem 4.34 (1 is a peripheral eigenvalue). *If E has a nonzero fixed point $\rho \neq 0$ with $E(\rho) = \rho$, then $1 \in \text{peripheral}(E)$.*

Proof. ρ is an eigenvector with eigenvalue 1, and $|1| = 1$ equals the spectral radius. $\square$

Theorem 4.35 (Peripheral eigenvalues lie on the unit circle). *Every peripheral eigenvalue λ satisfies $|\lambda| = 1$.*

Proof. By definition of peripheral eigenvalues (spectral radius = 1 for channels). $\square$

Theorem 4.36 (Finiteness of peripheral eigenvalues). *On a finite-dimensional space, the set of peripheral eigenvalues is finite.*

Proof. Peripheral eigenvalues are a subset of all eigenvalues, which is a finite set in finite dimensions. $\square$

Theorem 4.37 (Peripheral eigenvalues are roots of unity). *Every peripheral eigenvalue of a channel is a root of unity. By the multiplicative domain theory ([Wol12, Prop. 6.2]), if μ is a peripheral eigenvalue then μ^n is an eigenvalue for every $n \geq 1$. In finite dimensions the eigenvalue set is finite, so $\{\mu^n\}$ is finite, forcing μ to be a root of unity.*

Proof. The hypothesis of the Lean theorem is that μ^n is an eigenvalue for every n . By the pigeonhole principle on the finite eigenvalue set, $\mu^a = \mu^b$ for some $a \neq b$, giving $\mu^{|a-b|} = 1$. $\square$

Theorem 4.38 (Peripheral powers imply root of unity). *If the set of peripheral eigenvalues of a linear endomorphism E on a finite-dimensional space is closed under powers ($\mu \in \text{peripheral}(E)$ and $n \geq 1$ imply $\mu^n \in \text{peripheral}(E)$), then every peripheral eigenvalue is a root of unity.*

Proof. The closure hypothesis provides $\mu^n \in \text{peripheral}(E)$ for all n , which in particular means μ^n is an eigenvalue. Theorem 4.37 then gives $\mu^p = 1$ for some $p > 0$. $\square$

4.7.1 Periodicity removal by powering

Remark 4.39. The results in this subsection are purely spectral: they use only finiteness of a set of complex numbers and the spectral mapping theorem for powers. In particular, they do not rely on complete positivity.

Lemma 4.40 (Common exponent for a finite set of roots of unity). *Let $s \subseteq \mathbb{C}$ be a finite set. Assume that for every $\mu \in s$ there exists an exponent $p_\mu > 0$ with $\mu^{p_\mu} = 1$. Then there exists $p > 0$ such that $\mu^p = 1$ for all $\mu \in s$.*

Proof. Take p to be the least common multiple of the exponents p_μ . $\square$

Lemma 4.41 (Powering collapses peripheral eigenvalues). *Let $E : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ be a linear map, and assume that E has a nonzero fixed point $\rho \neq 0$. Let $p > 0$. If every peripheral eigenvalue μ of E satisfies $\mu^p = 1$, then $\text{peripheral}(E^p) = \{1\}$.*

Proof. The fixed point ρ gives $1 \in \text{peripheral}(E^p)$. For the reverse inclusion, let $\nu \in \text{peripheral}(E^p)$. Spectral mapping gives $\nu = \mu^p$ for some $\mu \in \sigma(E)$. From $|\nu| = 1$ and $p > 0$ we obtain $|\mu| = 1$, hence $\mu \in \text{peripheral}(E)$. The hypothesis $\mu^p = 1$ then forces $\nu = 1$. $\square$

Lemma 4.42 (Peripheral eigenvalues closed under powers). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is both unital and trace-preserving, and assume that E is irreducible. Then $\text{peripheral}(E)$ is closed under powers: if $\mu \in \text{peripheral}(E)$ then $\mu^n \in \text{peripheral}(E)$ for every $n \in \mathbb{N}$.*

Proof. Let $E(X) = \mu X$ with $|\mu| = 1$. The Kadison–Schwarz equality for peripheral eigenvectors (Theorem 4.20) shows that $X^\dagger X$ is a positive semidefinite fixed point. By irreducibility, this fixed point is positive definite (Theorem 5.2), hence X is invertible. Multiplicative-domain powers then give $E(X^n) = \mu^n X^n$, so μ^n is again peripheral. $\square$

Theorem 4.43 (Roots of unity for irreducible doubly stochastic maps). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is both unital and trace-preserving, and assume that E is irreducible. Then every peripheral eigenvalue μ of E is a root of unity: there exists $p > 0$ with $\mu^p = 1$.*

Proof. Combine the closure-under-powers result (Lemma 4.42) with Theorem 4.38. $\square$

4.7.2 Peripheral closure via adjoint fixed point

The doubly-stochastic hypothesis in Lemma 4.42 (unital and trace-preserving) can be weakened: trace preservation is only used to derive the Kadison–Schwarz equality at peripheral eigenvectors. The following result replaces the TP condition by the existence of a positive definite fixed point for the adjoint Kraus map, matching the weighted-trace argument in [CPGSV16, Appendix A].

Lemma 4.44 (Peripheral closure via adjoint fixed point). *Let $E(X) = \sum_i K_i X K_i^\dagger$ be a Kraus map that is unital ($\sum_i K_i K_i^\dagger = \mathbb{1}$), and assume:*

1. *the adjoint Kraus map $E^*(X) = \sum_i K_i^\dagger X K_i$ has a positive definite fixed point $\rho > 0$, and*

2. E is irreducible.

Then $\text{peripheral}(E)$ is closed under powers: if $\mu \in \text{peripheral}(E)$ then $\mu^n \in \text{peripheral}(E)$ for every $n \in \mathbb{N}$.

Proof. Let $E(X) = \mu X$ with $|\mu| = 1$. The weighted Kadison–Schwarz equality (using the adjoint fixed point ρ in place of trace preservation) shows $E(X^\dagger X) = E(X)^\dagger E(X)$. Hence $X^\dagger X$ is a PSD fixed point, which is positive definite by irreducibility (Theorem 5.2). So X is invertible, and multiplicative-domain powers give $E(X^n) = \mu^n X^n$, so μ^n is peripheral. $\square$

Theorem 4.45 (Roots of unity via adjoint fixed point). *Under the same hypotheses as Lemma 4.44, every peripheral eigenvalue of E is a root of unity.*

Proof. Combine the closure-under-powers result (Lemma 4.44) with Theorem 4.38. $\square$

Remark 4.46. Lemma 4.44 and Theorem 4.45 strictly generalise Lemma 4.42 and Theorem 4.43: the doubly-stochastic (unital + TP) condition implies $E^*(\mathbb{1}) = \mathbb{1}$, which is a positive definite fixed point of the adjoint. The adjoint-fixed-point formulation matches the argument in [CPGSV16, Appendix A], which uses a faithful invariant state rather than double stochasticity.

Definition 4.47 (Channel period). The *period* of a channel E is the number of peripheral eigenvalues (counted without multiplicity): $p(E) := |\text{peripheral}(E)|$.

Definition 4.48 (Primitive map). A linear map is *primitive* if its only peripheral eigenvalue is $\lambda = 1$, i.e. $\text{peripheral}(E) = \{1\}$. For channels this corresponds to [Wol12, Thm. 6.11]. The Lean definition applies to any linear endomorphism, not only channels.

Theorem 4.49 (Primitivity equals period one). *A linear map with a nonzero fixed point is primitive if and only if its period is 1.*

Proof. If $\text{peripheral}(E) = \{1\}$, then $|\text{peripheral}(E)| = 1$. Conversely, if $|\text{peripheral}(E)| = 1$, then since $1 \in \text{peripheral}(E)$ (Theorem 4.34), the only element is 1. $\square$

Theorem 4.50 (Non-peripheral eigenvalues of primitive maps). *If a linear map is primitive, then every eigenvalue $\lambda \neq 1$ satisfies $|\lambda| < 1$.*

Proof. By primitivity, no eigenvalue other than 1 has modulus 1. Since the spectral radius is 1, all other eigenvalues have strictly smaller modulus. $\square$

Chapter 5

Quantum Perron–Frobenius Theory

This chapter establishes that the transfer map of an injective MPS tensor has a unique positive semidefinite fixed point (up to scaling), and that this fixed point is positive definite. The results follow [EHK78] and [Wol12, Ch. 6].

5.1 Positive definiteness

Theorem 5.1 (PSD fixed point is positive definite). *Let A be an injective MPS tensor and let $\rho \geq 0$, $\rho \neq 0$, be a fixed point of the transfer map $\mathcal{E}_A(\rho) = \rho$. Then ρ is positive definite.*

Proof. Suppose $v^\dagger \rho v = 0$ for some $v \neq 0$. Since $\rho \geq 0$, this implies $\rho v = 0$. The fixed-point equation gives $v^\dagger \rho v = \sum_i ((A^i)^\dagger v)^\dagger \rho ((A^i)^\dagger v)$. Each term is nonneg; if the sum is zero, then $\rho (A^i)^\dagger v = 0$ for all i : the kernel of ρ is invariant under all $(A^i)^\dagger$. Since the $\{A^i\}$ span $M_D(\mathbb{C})$, $\rho M v = 0$ for all M , which forces $\rho = 0$, contradicting $\rho \neq 0$. In [Wol12, Thm. 6.8], positive definiteness is proved using the growth condition $(\mathbb{1} + \mathcal{E})^{D-1}(X) > 0$ for irreducible maps ([Wol12, Thm. 6.6, condition (2)]). Our proof exploits injectivity directly, giving a shorter argument. $\square$

Theorem 5.2 (PSD fixed point is PD — irreducible version). *If $\mathcal{E}_A$ is irreducible and $\rho \geq 0$, $\rho \neq 0$ is a fixed point, then ρ is positive definite.*

Proof. Irreducibility of a Kraus map implies the hypotheses of Theorem 5.1, so the result follows. Alternatively, one can use the irreducibility growth chain $(\mathbb{1} + \mathcal{E})^{D-1}(X) > 0$ from [Wol12, Thm. 6.6] directly. $\square$

5.2 Uniqueness

Theorem 5.3 (Uniqueness of PSD fixed point). *Let A be an injective MPS tensor. If $\rho, \sigma \geq 0$ are both nonzero fixed points of $\mathcal{E}_A$, then $\sigma = c\rho$ for some $c \in \mathbb{C}$.*

Proof. Both ρ and σ are positive definite by Theorem 5.1. Write $\rho = SS^\dagger$ via a spectral square root and set $H = (S^\dagger)^{-1} \sigma S^{-1}$, which is Hermitian and positive definite. Let c_0 be the minimal eigenvalue of H . Then $\tau := \sigma - c_0 \rho$ is a PSD fixed point which is *not* positive definite (it has a zero eigenvalue by construction of c_0). By Theorem 5.1, τ must be zero, giving $\sigma = c_0 \rho$.

In [Wol12, Thm. 6.8], uniqueness follows from non-degeneracy of the spectral radius eigenvalue via the $(\mathbb{1} + \mathcal{E})^{D-1}$ growth condition. Our proof reduces uniqueness to positive definiteness via a spectral shift argument. $\square$

5.3 Existence and assembly

Theorem 5.4 (Existence of PSD fixed point). *Let A be an injective MPS tensor with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ (so that $\mathcal{E}_A$ is trace-preserving). Then there exists $\rho \geq 0$, $\rho \neq 0$, with $\mathcal{E}_A(\rho) = \rho$.*

Proof. Under the normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$, the transfer map is a quantum channel (Theorem 4.25). The Cesàro fixed-point theorem (Theorem 4.29) provides the fixed point. $\square$

Theorem 5.5 (Quantum Perron–Frobenius). *Let A be an injective MPS tensor (with $\sum_i (A^i)^\dagger A^i = \mathbb{1}$). Then the transfer map $\mathcal{E}_A$ has a unique PSD fixed point ρ (up to scaling), and ρ is positive definite.*

Proof. Combine Theorems 5.4, 5.1, and 5.3. $\square$

Theorem 5.6 (QPF without dimension hypothesis). *The conclusion of Theorem 5.5 holds without the hypothesis $D \geq 1$: when $D = 0$, the zero matrix is vacuously the unique positive definite fixed point.*

Proof. For $D \geq 1$, apply Theorem 5.5. For $D = 0$, all 0×0 matrices are equal. $\square$

5.4 Doubly stochastic gauge

Theorem 5.7 (Unital gauge). *Let $\mathcal{E}_A(\rho) = \rho$ and let S be an invertible matrix with $SS^\dagger = \rho$. Define the gauged operators $A'^i = S^{-1}A^iS$. Then $\sum_i A'^i(A'^i)^\dagger = \mathbb{1}$, i.e. the gauged Kraus map is unital.*

Proof. Expand each summand: $(S^{-1}A^iS)(S^{-1}A^iS)^\dagger = S^{-1}A^i \underbrace{SS^\dagger}_\rho (A^i)^\dagger (S^\dagger)^{-1}$. Summing over i and using $\sum_i A^i(A^i)^\dagger = \rho$ gives $S^{-1}\rho(S^\dagger)^{-1} = S^{-1}SS^\dagger(S^\dagger)^{-1} = \mathbb{1}$. $\square$

Theorem 5.8 (Trace-preserving gauge). *Let σ be a fixed point of the adjoint transfer map $\sum_i (A^i)^\dagger \sigma A^i = \sigma$, and let S be invertible with $S^\dagger S = \sigma$. Define $A'^i = SA^iS^{-1}$. Then $\sum_i (A'^i)^\dagger A'^i = \mathbb{1}$, i.e. the gauged Kraus map is trace-preserving.*

Proof. Expand each summand: $(SA^iS^{-1})^\dagger (SA^iS^{-1}) = (S^\dagger)^{-1}(A^i)^\dagger \underbrace{S^\dagger S}_\sigma A^i S^{-1}$. Summing over i and using $\sum_i (A^i)^\dagger \sigma A^i = \sigma$ gives $(S^\dagger)^{-1}\sigma S^{-1} = (S^\dagger)^{-1}S^\dagger S S^{-1} = \mathbb{1}$. $\square$

Chapter 6

Spectral Gap and Block Separation

This chapter proves that the MPV overlaps between distinct (non-gauge-phase-equivalent) MPS blocks decay to zero as system size grows, while self-overlaps of primitive blocks converge to 1. The argument is based on a spectral analysis of the *mixed transfer operator*, following [PGVWC07].

Throughout this chapter, all MPS tensors are assumed to satisfy the trace-preserving normalization $\sum_i (A^i)^\dagger A^i = \mathbb{1}$.

6.1 Mixed transfer operator

Definition 6.1 (Mixed transfer operator). The *mixed transfer operator* (or *cross transfer operator*) for two MPS tensors A and B of the same bond dimension D is the linear map $F_{AB} : M_D(\mathbb{C}) \rightarrow M_D(\mathbb{C})$ defined by

$$F_{AB}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Definition 6.2 (Rectangular mixed transfer operator). For MPS tensors A of bond dimension D_1 and B of bond dimension D_2 (possibly $D_1 \neq D_2$), the *rectangular mixed transfer operator* is $F_{AB}^{\text{rect}} : M_{D_1 \times D_2}(\mathbb{C}) \rightarrow M_{D_1 \times D_2}(\mathbb{C})$ defined by

$$F_{AB}^{\text{rect}}(X) = \sum_{i=0}^{d-1} A^i X (B^i)^\dagger.$$

Theorem 6.3 (Self-transfer). $F_{AA} = \mathcal{E}_A$.

Proof. Immediate from the definitions. □

Theorem 6.4 (Iterated mixed transfer). For all $X \in M_D(\mathbb{C})$ and $N \geq 0$,

$$F_{AB}^N(X) = \sum_{\sigma \in \{0, \dots, d-1\}^N} A^\sigma X (B^\sigma)^\dagger,$$

where $A^\sigma = A^{\sigma_1} \dots A^{\sigma_N}$ denotes the word evaluation (Definition 2.2).

Proof. Induction on N : the base case is $F^0(X) = X$; the inductive step applies F_{AB} to each summand and re-indexes via $\{0, \dots, d-1\}^{N+1} \cong \{0, \dots, d-1\} \times \{0, \dots, d-1\}^N$. $\square$

Theorem 6.5 (Matrix trace of iterated mixed transfer at the identity). *For $N \geq 0$,*

$$\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^{\sigma}(B^{\sigma})^{\dagger}).$$

This is the matrix trace of F_{AB}^N evaluated at the identity matrix. It should not be confused with the operator trace $\text{Tr}(F_{AB}^N)$, which equals the MPV overlap (Theorem 6.7).

Proof. Apply the matrix trace to each summand in Theorem 6.4 with $X = \mathbb{1}$: $\text{tr}(F_{AB}^N(\mathbb{1})) = \sum_{\sigma} \text{tr}(A^{\sigma} \cdot \mathbb{1} \cdot (B^{\sigma})^{\dagger}) = \sum_{\sigma} \text{tr}(A^{\sigma}(B^{\sigma})^{\dagger})$. $\square$

6.2 MPV overlap as transfer trace

Lemma 6.6 (Trace expansion over matrix units). *For any linear endomorphism T on $M_D(\mathbb{C})$, the operator trace can be expanded as*

$$(\text{tr} \circ T) = \sum_{p,q=0}^{D-1} (T(E_{pq}))_{pq},$$

where E_{pq} is the matrix with 1 in position (p, q) and 0 elsewhere.

Proof. The operator trace of T equals the sum of $\text{tr}(T(E_{pq}) \cdot E_{qp})$ over all (p, q) , and $\text{tr}(M \cdot E_{qp}) = M_{pq}$. $\square$

Theorem 6.7 (Overlap equals transfer trace). *For normalized tensors A, B of bond dimension $D \geq 1$,*

$$O_{AB}(N) = \text{tr}(F_{AB}^N),$$

where the right-hand side is the operator trace (trace of F_{AB}^N as a linear map on $M_D(\mathbb{C})$).

Proof. Expand the operator trace using Lemma 6.6 as a sum over matrix units E_{pq} . Apply Theorem 6.4 to evaluate $F_{AB}^N(E_{pq})$, extract the (p, q) -entry, and reassemble. The resulting double sum over (p, q) and σ equals $\sum_{\sigma} V^{(N)}(A)_{\sigma} \overline{V^{(N)}(B)_{\sigma}} = O_{AB}(N)$. $\square$

Theorem 6.8 (Rectangular overlap as transfer trace). *For tensors A of bond dimension $D_1 \geq 1$ and B of bond dimension $D_2 \geq 1$, $O_{AB}(N) = \text{tr}((F_{AB}^{\text{rect}})^N)$.*

Proof. Same argument as Theorem 6.7, using rectangular matrix units. $\square$

Theorem 6.9 (Inner product via transfer trace). $\langle V^{(N)}(A) | V^{(N)}(B) \rangle = \text{tr}(F_{AB}^N(\mathbb{1}))$.

Proof. Apply Theorem 6.4 with $X = \mathbb{1}$ and take the trace. $\square$

6.3 Eigenvalue bound and spectral gap

The proofs in this section use the Hilbert–Schmidt (Frobenius) norm contraction from the Kadison–Schwarz inequality (Theorem 4.17) and the doubly stochastic gauge (Theorem 5.7).

Theorem 6.10 (Eigenvalue bound). *Let A, B be normalized MPS tensors. Every eigenvalue μ of F_{AB} satisfies $|\mu| \leq 1$.*

Proof. Let $F_{AB}(X) = \mu X$ with $X \neq 0$. After passing to the doubly stochastic gauge (Theorem 5.7), the Hilbert–Schmidt contraction (Theorem 4.17) gives $\|F_{AB}^n(X)\|_{\text{HS}}^2 \leq \|X\|_{\text{HS}}^2$ for all n . Since $F_{AB}^n(X) = \mu^n X$, this gives $|\mu|^{2n} \leq D^2/\|X\|_{\text{HS}}^2$ (using $\|X\|_{\text{HS}}^2 \leq D^2$ for matrices in $M_D(\mathbb{C})$). If $|\mu| > 1$, the left-hand side diverges, contradicting the bound.

This proof avoids the Perron–Frobenius theorem entirely; the bound follows from the Kadison–Schwarz inequality alone. $\square$

Theorem 6.11 (Spectral radius bound). *$\rho(F_{AB}) \leq 1$ for normalized tensors.*

Proof. The spectral radius is the supremum of $|\mu|$ over all eigenvalues, and each satisfies $|\mu| \leq 1$ by Theorem 6.10. $\square$

Theorem 6.12 (Modulus-one eigenvalue implies gauge-phase equivalence). *Let A, B be injective normalized MPS tensors. If $\rho(F_{AB}) \geq 1$, then A and B are gauge-phase equivalent.*

Proof. Since $\rho(F_{AB}) \leq 1$ (Theorem 6.11) and $\rho(F_{AB}) \geq 1$ by hypothesis, there exists an eigenvalue μ with $|\mu| = 1$ and eigenvector $X \neq 0$: $F_{AB}(X) = \mu X$.

After passing to the doubly stochastic gauge, F_{AB} becomes both unital and trace-preserving. By the KS equality for peripheral eigenvectors (Theorem 4.20), the Kadison–Schwarz gap vanishes. The Kraus commutation identity (Theorem 4.19) then gives $X(B^i)^\dagger = (B^i)^\dagger \mu X$ for all i . Using the multiplicative domain (Theorem 4.21), one shows that X is invertible (using injectivity of B and simplicity of $M_D(\mathbb{C})$, Theorem 3.10). Skolem–Noether (Theorem 3.11) then yields $B^i = \bar{\mu} X A^i X^{-1}$ for all i , establishing gauge-phase equivalence (Definition 2.13).

In [PGVWC07], this argument appears as part of the proof of the single-block fundamental theorem. Our formalization factors it out as a self-contained spectral-gap theorem for the mixed transfer operator, following the same chain (KS inequality $\rightarrow$ multiplicative domain $\rightarrow$ Skolem–Noether). $\square$

Theorem 6.13 (Strict spectral gap). *Let A, B be injective normalized MPS tensors that are not gauge-phase equivalent. Then $\rho(F_{AB}) < 1$.*

Proof. $\rho(F_{AB}) \leq 1$ by Theorem 6.11. If $\rho(F_{AB}) = 1$, Theorem 6.12 gives gauge-phase equivalence, contradicting the hypothesis. $\square$

6.4 Spectral gap — rectangular case

Theorem 6.14 (Rectangular spectral gap). *Let A be an injective normalized tensor of bond dimension D_1 and B an injective normalized tensor of bond dimension D_2 , with $D_1 \neq D_2$. Then $\rho(F_{AB}^{\text{rect}}) < 1$.*

Proof. The eigenvalue bound $|\mu| \leq 1$ holds by the same Frobenius-norm contraction argument as Theorem 6.10, adapted to rectangular matrices. If $\rho = 1$, the modulus-one eigenvector argument yields an invertible intertwiner $X : \mathbb{C}^{D_2} \rightarrow \mathbb{C}^{D_1}$, which is impossible when $D_1 \neq D_2$.

The rectangular case does not appear in [PGVWC07]; the formalization extends the square-case argument by observing that the dimension mismatch obstructs the final Skolem–Noether step. $\square$

Theorem 6.15 (Rectangular overlap decay). *If $D_1 \neq D_2$ and both tensors are injective and normalized, then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. The spectral gap $\rho(F_{AB}^{\text{rect}}) < 1$ (Theorem 6.14) implies $(F_{AB}^{\text{rect}})^N \rightarrow 0$, so the operator trace converges to 0. By Theorem 6.8, $O_{AB}(N) = \text{tr}((F_{AB}^{\text{rect}})^N) \rightarrow 0$. $\square$

6.5 MPV overlap decay

Theorem 6.16 (Transfer powers converge to zero). *Let A, B be injective normalized tensors that are not gauge-phase equivalent. Then $F_{AB}^n(X) \rightarrow 0$ for every $X \in M_D(\mathbb{C})$.*

Proof. The spectral gap $\rho(F_{AB}) < 1$ (Theorem 6.13) implies $\|F_{AB}^n\|_{\text{op}} \rightarrow 0$ by the Gelfand formula, so $F_{AB}^n(X) \rightarrow 0$ for every X . $\square$

Theorem 6.17 (Overlap decay). *Let A, B be injective normalized tensors of the same bond dimension that are not gauge-phase equivalent. Then $O_{AB}(N) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. By Theorem 6.7, $O_{AB}(N) = \text{tr}(F_{AB}^N)$. Expand the operator trace over matrix units (Lemma 6.6): $\text{tr}(F_{AB}^N) = \sum_{p,q} (F_{AB}^N(E_{pq}))_{pq}$. By Theorem 6.16, each $F_{AB}^N(E_{pq}) \rightarrow 0$ entry-wise. Since the sum is finite, $O_{AB}(N) \rightarrow 0$. $\square$

6.6 Block separation

Theorem 6.18 (Cross-correlation decay). *Let A, B be injective normalized tensors that are not gauge-phase equivalent. For every $X \in M_D(\mathbb{C})$, $\text{tr}(F_{AB}^N(X)) \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. By Theorem 6.16, $F_{AB}^N(X) \rightarrow 0$; the trace is continuous in finite dimensions. $\square$

Theorem 6.19 (Self-correlation persists). *Let ρ be the unique PSD fixed point of $\mathcal{E}_A$ (Theorem 5.5). Then $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$ for all $N \geq 0$.*

Proof. Since $\mathcal{E}_A(\rho) = \rho$, by induction $\mathcal{E}_A^N(\rho) = \rho$, so $\text{tr}(\mathcal{E}_A^N(\rho)) = \text{tr}(\rho)$. $\square$

Theorem 6.20 (Block separation principle). *Let A, B be injective tensors that are not gauge-phase equivalent. If $\text{tr}(F_{AB}^N(\mathbb{1})) = 0$ for all $N \geq 0$, then $F_{AB}(\mathbb{1}) = 0$.*

Proof. Taking $N = 0$ gives $\text{tr}(F_{AB}^0(\mathbb{1})) = \text{tr}(\mathbb{1}) = D$. For $D \geq 1$, this is nonzero, so the hypothesis is vacuously false. For $D = 0$, all 0×0 matrices are equal. $\square$

6.7 Primitive overlap convergence

Theorem 6.21 (Trace of powers converges to 1). *Let E be trace-preserving with fixed point ρ ($\text{tr}(\rho) \neq 0$) and let P be the fixed-point projection (Definition 4.30). If $\rho(E - P) < 1$, then $\text{tr}(E^n) \rightarrow 1$ as $n \rightarrow \infty$.*

Proof. By the power decomposition (Theorem 4.31), $E^n = P + (E - P)^n$ for $n \geq 1$. Since $\rho(E - P) < 1$, the Gelfand formula gives $(E - P)^n \rightarrow 0$, so $\text{tr}(E^n) \rightarrow \text{tr}(P)$. The operator trace of P equals 1: since $P(X) = \frac{\text{tr}(X)}{\text{tr}(\rho)}\rho$, we have $\text{tr}(P) = \frac{\text{tr}(\rho)}{\text{tr}(\rho)} = 1$. $\square$

Theorem 6.22 (Primitive overlap convergence). *Let A be a normalized MPS tensor with a nonzero PSD fixed point ρ of the transfer map. If $\rho(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection), then $O_{AA}(N) \rightarrow 1$ as $N \rightarrow \infty$.*

Proof. By Theorem 6.7 (with $A = B$), $O_{AA}(N) = \text{tr}(\mathcal{E}_A^N)$. By Theorem 6.21, $\text{tr}(\mathcal{E}_A^N) \rightarrow 1$. $\square$

Chapter 7

Wielandt Bound

This chapter proves a quantum analog of Wielandt's inequality: for a normal MPS tensor with bond dimension D , the cumulative span of matrix products stabilises after at most D^2 steps. The results follow [SPGWC10].

7.1 Cumulative span

Definition 7.1 (Word span). The *word span* at length n is the subspace

$$S_n(A) = \text{span}_{\mathbb{C}}\{A^w : |w| = n\} \subseteq M_D(\mathbb{C}).$$

This is [SPGWC10, Eq. (1)].

Definition 7.2 (Cumulative span). The *cumulative span* at level n is

$$T_n(A) = S_0(A) + S_1(A) + \cdots + S_n(A).$$

Lemma 7.3 (Monotonicity of cumulative span). $T_n(A) \leq T_{n+1}(A)$ for all n .

Proof. T_{n+1} includes all words of length $\leq n+1$, which contains all words of length $\leq n$. $\square$

Lemma 7.4 (Dimension bound). $\dim T_n(A) \leq D^2$ for all n .

Proof. $T_n(A) \subseteq M_D(\mathbb{C})$ and $\dim M_D(\mathbb{C}) = D^2$. $\square$

Theorem 7.5 (Cumulative span stabilisation). If $T_n(A) = T_{n+1}(A)$, then $T_m(A) = T_n(A)$ for all $m \geq n$.

Proof. If $T_n = T_{n+1}$, then $S_{n+1} \subseteq T_n$. Left-multiplying any element of S_{n+1} by A^i gives an element of S_{n+2} , so $S_{n+2} \subseteq T_{n+1} = T_n$. By induction, $S_m \subseteq T_n$ for all $m > n$, hence $T_m = T_n$. $\square$

Theorem 7.6 (Dimension growth). If $T_n(A) \neq T_{n+1}(A)$, then $\dim T_{n+1}(A) > \dim T_n(A)$.

Proof. $T_n \leq T_{n+1}$ (Lemma 7.3) and $T_n \neq T_{n+1}$ imply strict submodule inclusion, hence strict rank increase. $\square$

Lemma 7.7 (Word span characterises block injectivity). $S_L(A) = M_D(\mathbb{C})$ if and only if A is L -block injective.

Proof. Both conditions assert that the span of all products of length L equals $M_D(\mathbb{C})$. $\square$

7.2 Fitting decomposition

Definition 7.8 (Fitting decomposition). A *Fitting decomposition* of a linear endomorphism $f : V \rightarrow V$ on a finite-dimensional vector space over an algebraically closed field consists of:

1. f is nilpotent on the generalized 0-eigenspace,
2. f is invertible on each generalized μ -eigenspace for $\mu \neq 0$,
3. the generalized eigenspaces span V ,
4. the generalized eigenspaces are linearly independent.

Theorem 7.9 (Fitting decomposition exists). *Every linear endomorphism on a finite-dimensional vector space over an algebraically closed field admits a Fitting decomposition.*

Proof. Nilpotency on the zero generalized eigenspace follows from the definition of generalized eigenspaces. Invertibility on nonzero generalized eigenspaces follows because $f - \mu$ is nilpotent there, so $f = \mu(1 - (1 - f/\mu))$ is invertible. Spanning and independence are standard results for generalized eigenspaces over algebraically closed fields. In [SPGWC10] and [Wol12, Lemma 6.3(b)], the Jordan normal form is used directly. Our formalization uses generalized eigenspaces as a substitute, since the full Jordan normal form is not available in Mathlib. $\square$

Theorem 7.10 (Nilpotent power bound). *If f is nilpotent on a space of dimension n , then $f^n = 0$.*

Proof. A nilpotent endomorphism on an n -dimensional space has nilpotency index $\leq n$. $\square$

7.3 Nonzero trace product

Theorem 7.11 (Cumulative span reaches $M_D(\mathbb{C})$ — explicit bound). *If A is normal, then there exists $n \leq D^2$ with $T_n(A) = M_D(\mathbb{C})$.*

Proof. By Theorem 7.6, if $T_n \neq T_{n+1}$ then $\dim T_{n+1} > \dim T_n$. Since $\dim T_n \leq D^2$ (Lemma 7.4), after at most D^2 steps either $T_n = M_D(\mathbb{C})$ or T_n has stabilised. Stabilisation before reaching $M_D(\mathbb{C})$ contradicts normality (which requires $S_{L_0} = M_D(\mathbb{C})$ for some L_0 , hence $T_{L_0} = M_D(\mathbb{C})$). $\square$

Theorem 7.12 (Cumulative span reaches $M_D(\mathbb{C})$). *If A is normal, then $T_{D^2}(A) = M_D(\mathbb{C})$.*

Proof. Take n from Theorem 7.11; since $n \leq D^2$ and T is monotone, $T_{D^2} = M_D(\mathbb{C})$. $\square$

Theorem 7.13 (Nonzero trace product). *If A is normal, then there exists a word w of length $\leq D^2$ such that $\text{tr}(A^w) \neq 0$. This is [SPGWC10, Lemma 1].*

Proof. By Theorem 7.12, $\mathbb{1} \in T_{D^2}(A)$, so $\mathbb{1}$ is a linear combination of word products of length $\leq D^2$. At least one has nonzero trace (since $\text{tr}(\mathbb{1}) = D \neq 0$). $\square$

7.4 Eigenvector extraction

Theorem 7.14 (Nonzero trace implies eigenvector). *If $M \in M_D(\mathbb{C})$ has $\text{tr}(M) \neq 0$, then M has a nonzero eigenvalue $\mu \neq 0$ and a corresponding eigenvector $\varphi \neq 0$ with $M\varphi = \mu\varphi$.*

Proof. A matrix with nonzero trace has a nonzero eigenvalue (since trace = sum of eigenvalues counted with multiplicity). The Fitting decomposition (Theorem 7.9) ensures the corresponding generalized eigenspace is nontrivial, yielding an eigenvector. $\square$

Theorem 7.15 (Eigenvalue extraction). *If A is normal, then there exist a word w_0 with $|w_0| \leq D^2$, a nonzero scalar $\mu \neq 0$, and a nonzero vector $\varphi \neq 0$ such that $A^{w_0}\varphi = \mu\varphi$.*

Proof. By Theorem 7.13, there is a word w_0 with $\text{tr}(A^{w_0}) \neq 0$ and $|w_0| \leq D^2$. By Theorem 7.14, A^{w_0} has a nonzero eigenvector. $\square$

7.5 Eigenvector spreading

Definition 7.16 (Cumulative vector span). Given an MPS tensor A and a vector $\varphi \in \mathbb{C}^D$, the *cumulative vector span* is

$$K_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w\varphi : |w| \leq n\} \subseteq \mathbb{C}^D.$$

This is [SPGWC10, proof of Lemma 2(a)].

Theorem 7.17 (Vector span stabilisation). *If $K_n(A, \varphi) = K_{n+1}(A, \varphi)$, then $K_m(A, \varphi) = K_n(A, \varphi)$ for all $m \geq n$.*

Proof. Same argument as Theorem 7.5: left-multiplication by A^i cannot escape the stabilised span. $\square$

Lemma 7.18 (Vector span from matrix span). *If $T_N(A) = M_D(\mathbb{C})$ and $\varphi \neq 0$, then $K_N(A, \varphi) = \mathbb{C}^D$.*

Proof. Every matrix $M \in M_D(\mathbb{C})$ is in $T_N(A)$, so in particular every rank-one matrix sending φ to any target vector v is in T_N . Applying such matrices to φ recovers all of $\mathbb{C}^D$. $\square$

Theorem 7.19 (Eigenvector spreading). *Let A be a normal MPS tensor and let $A^{i_0}\varphi = \mu\varphi$ with $\mu \neq 0$ and $\varphi \neq 0$. Then $K_{D-1}(A, \varphi) = \mathbb{C}^D$. This is [SPGWC10, Lemma 2(a)].*

Proof. The vector span K_n is monotone and $\dim K_n \leq D$. If $K_n = K_{n+1}$ for some $n < D-1$, the span stabilises (Theorem 7.17). But $T_{D^2}(A) = M_D(\mathbb{C})$ by Theorem 7.12, so Lemma 7.18 gives $K_{D^2}(A, \varphi) = \mathbb{C}^D$, contradicting early stabilisation. Hence $\dim K_n$ grows by ≥ 1 at each step, reaching D by step $D-1$. $\square$

7.6 Assembly

Definition 7.20 (Wielandt analysis). A *Wielandt analysis* for a normal MPS tensor A packages: a word w_0 of length $\leq D^2$, a nonzero eigenvalue μ , and a nonzero eigenvector φ of the matrix A^{w_0} , such that $A^{w_0}\varphi = \mu\varphi$.

Theorem 7.21 (Wielandt analysis exists). *Every normal MPS tensor admits a Wielandt analysis.*

Proof. Immediate from Theorem 7.15. $\square$

Theorem 7.22 (Wielandt chain). *Every normal MPS tensor admits a Wielandt analysis (Theorem 7.21), and the eigenvector from that analysis spreads to all of $\mathbb{C}^D$ in $D - 1$ steps (Theorem 7.19).*

Proof. Combine Theorems 7.21 and 7.19. $\square$

Theorem 7.23 (Wielandt bound). *If A is a normal MPS tensor with bond dimension D , then $T_{D^2}(A) = M_D(\mathbb{C})$. In particular, the cumulative span stabilises at index at most D^2 . This is [SPGWC10, Theorem 1] (which gives the sharper bound $D^2 - d + 1$ in certain cases; our formalization proves the D^2 bound).*

Proof. Immediate from Theorem 7.12. $\square$

7.7 Lemma 2(b): fixed-length matrix spanning

The preceding sections establish Lemma 2(a) of [SPGWC10]: eigenvector spreading gives a cumulative vector span $K_{D-1}(A, \varphi) = \mathbb{C}^D$. The next step, Lemma 2(b), strengthens this to *fixed-length* matrix spanning: there exists an n_0 such that $S_{n_0}(A) = M_D(\mathbb{C})$ (i.e. word products of *exactly* one length span all matrices). The formalization in `Wielandt/Lemma2b.lean` covers the algebraic assembly part of this argument.

Definition 7.24 (Fixed-length vector span). The *fixed-length vector span* at length n is

$$H_n(A, \varphi) = \text{span}_{\mathbb{C}}\{A^w \varphi : |w| = n\} \subseteq \mathbb{C}^D.$$

This is $S_n(A)|\varphi\rangle$ in the notation of [SPGWC10].

Lemma 7.25 (Word span products). $S_m(A) \cdot S_n(A) \subseteq S_{m+n}(A)$: *the product of word spans at lengths m and n is contained in the word span at length $m + n$.*

Proof. Every product $A^{w_1} A^{w_2}$ with $|w_1| = m$ and $|w_2| = n$ equals $A^{w_1 w_2}$ with $|w_1 w_2| = m + n$. $\square$

Lemma 7.26 (Eigenvector padding). *Suppose $A^{i_0} \varphi = \mu \varphi$ with $\mu \neq 0$. Then for all n ,*

$$K_n(A, \varphi) = H_n(A, \varphi).$$

In particular, if $K_n(A, \varphi) = \mathbb{C}^D$ then already $H_n(A, \varphi) = \mathbb{C}^D$.

Proof. Any word w with $|w| \leq n$ can be padded to exact length n by appending $n - |w|$ copies of the index i_0 . The eigenvector relation gives $A^{w \cdot i_0^k} \varphi = \mu^k \cdot A^w \varphi$, so the padded vector is a nonzero scalar multiple of the original. Hence every generator of K_n lies in H_n . The reverse inclusion $H_n \leq K_n$ is immediate. $\square$

Theorem 7.27 (Vector spanning to matrix spanning (assembly step)). *Assume:*

1. $H_n(A, \varphi) = \mathbb{C}^D$ (length- n word products applied to φ span all of $\mathbb{C}^D$), and
2. for each standard basis vector e_j , the rank-one operator $|\varphi\rangle\langle e_j|$ lies in $S_m(A)$.

Then $S_{n+m}(A) = M_D(\mathbb{C})$.

Proof. Fix a matrix unit E_{ij} . By hypothesis (1), there exists $M_i \in S_n(A)$ with $M_i\varphi = e_i$. By hypothesis (2), $|\varphi\rangle\langle e_j| \in S_m(A)$. Their product satisfies

$$M_i \cdot |\varphi\rangle\langle e_j| = |M_i\varphi\rangle\langle e_j| = |e_i\rangle\langle e_j| = E_{ij},$$

and $E_{ij} \in S_{n+m}(A)$ by Lemma 7.25. Since the matrix units span $M_D(\mathbb{C})$, $S_{n+m}(A) = M_D(\mathbb{C})$. $\square$

Lemma 7.28 (Blocking transfer). *If the blocked tensor $A^{[L]}$ satisfies $S_n(A^{[L]}) = M_D(\mathbb{C})$, then $S_{nL}(A) = M_D(\mathbb{C})$.*

Proof. Each length- n word in the blocked alphabet $\{0, \dots, d^L - 1\}$ decodes to a length- nL word in the original alphabet $\{0, \dots, d - 1\}$, so $S_n(A^{[L]}) \subseteq S_{nL}(A)$. $\square$

Remark 7.29 (Status of Lemma 2(b)). Theorem 7.27 establishes the *assembly* part of Lemma 2(b) of [SPGWC10]: rank-one operators plus fixed-length vector spanning imply fixed-length matrix spanning. What remains is the *rank-one construction step*: showing that the Fitting/Jordan analysis of the eigenvalue word A^{w_0} (from Theorem 7.15) produces the required rank-one operators $|\varphi\rangle\langle e_j| \in S_m(A)$. Once this is done, combining with the eigenvector spreading (Theorem 7.19) and eigenvector padding (Lemma 7.26) will yield the full Lemma 2(b) and hence the sharp Wielandt bound $S_{(D-1)^2+1}(A) = M_D(\mathbb{C})$.

7.8 Blocked tensor and rectangular span

The module `Wielandt/RectangularSpan.lean` extends the Lemma 2(b) infrastructure to the blocked tensor $A^{[L]}$ and develops the “rectangular span” used to produce rank-one elements.

Theorem 7.30 (Blocking preserves normality). *If A is normal and $L > 0$, then the blocked tensor $A^{[L]}$ is also normal.*

Proof. If $S_N(A) = M_D(\mathbb{C})$ then $S_{NL}(A) = M_D(\mathbb{C})$ (padding by repeated single-letter words). By Lemma 7.31, $S_{NL}(A) \subseteq S_N(A^{[L]})$, so $S_N(A^{[L]}) = M_D(\mathbb{C})$. $\square$

Lemma 7.31 (Chunking: word span embeds into blocked word span). *$S_{nL}(A) \subseteq S_n(A^{[L]})$ for all n, L .*

Proof. Every length- nL word in the original alphabet can be split into n consecutive chunks of length L , each of which is a single blocked Kraus operator. By induction on n , using the word span product lemma (Lemma 7.25). $\square$

Theorem 7.32 (Word eigenvector gives blocked eigenvector). *If $A^{w_0}\varphi = \mu\varphi$ for a word w_0 of length L , then $(A^{[L]})^{j_0}\varphi = \mu\varphi$, where j_0 is the encoded blocked index corresponding to w_0 .*

Proof. By definition, $(A^{[L]})^{j_0} = A^{w_0}$. $\square$

Theorem 7.33 (Wielandt blocked assembly). *Suppose A is normal and $L > 0$, and we have:*

1. *a word w_0 of length L with eigenvector $A^{w_0}\varphi = \mu\varphi$ ($\mu \neq 0$, $\varphi \neq 0$),*
2. *a word τ_0 of length L with transpose eigenvector $(A^{\tau_0})^T\psi = \nu\psi$ ($\nu \neq 0$, $\psi \neq 0$),*
3. *a rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for some m .*

Then $S_{((D-1)+m+(D-1)) \cdot L}(A) = M_D(\mathbb{C})$.

Proof. Apply the conditional Lemma 2(b) assembly (Theorem 7.27) to the blocked tensor $A^{[L]}$, which is normal by Theorem 7.30. The eigenvector and transpose eigenvector are transferred to single-index eigenvectors of $A^{[L]}$ by Theorem 7.32. Transfer back to the original tensor via Lemma 7.28. $\square$

Remark 7.34 (Remaining gap for Lemma 2(b) via blocking). Theorem 7.33 reduces the full Lemma 2(b) of [SPGWC10] to producing a single rank-one element $\varphi\psi^T \in S_m(A^{[L]})$ for bounded m . This is the core linear-algebraic step: the Fitting/Jordan analysis of high powers of the eigenvector word products should produce this rank-one element via the “rectangular span” developed in the same module. The rectangular span API (monotonicity, dimension bounds, Wielandt-type stabilisation) is formalized, but the final rank-one extraction is not yet completed.

7.9 Primitivity and normality

Theorem 7.35 (Primitive MPS tensors have nonzero word products). *If A is a primitive MPS tensor (with DS gauge), then for every $n \geq 0$ there exists a word w of length exactly n with $A^w \neq 0$. (The Lean definition `IsPrimitive` bundles DS-gauge normalization; `IsNormal` is not required.)*

Proof. The PSD fixed point ρ of $\mathcal{E}_A$ satisfies $\mathcal{E}_A^n(\rho) = \rho \neq 0$. Since $\mathcal{E}_A^n(\rho) = \sum_{|w|=n} A^w \rho (A^w)^\dagger$, at least one summand is nonzero, hence some $A^w \neq 0$. $\square$

Theorem 7.36 (Iterated transfer does not vanish). *If A is a primitive MPS tensor (with DS gauge), then $\mathcal{E}_A^n \neq 0$ for all $n \geq 0$.*

Proof. By Theorem 7.35, there exists a word w of length n with $A^w \neq 0$, so $\mathcal{E}_A^n(\rho)$ has a nonzero summand. $\square$

7.9.1 Primitivity-to-normality bridge

The module `Wielandt/PrimitivityToNormal.lean` studies the relationship between the spectral-gap notion of primitivity (`IsPrimitiveMPS`: unique PSD fixed point with spectral gap) and normality.

Theorem 7.37 (Fixed-point uniqueness from spectral gap). *If A is a primitive MPS tensor with PSD fixed point ρ , then every fixed point σ of $\mathcal{E}_A$ is proportional to ρ : $\sigma = \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$.*

Proof. Set $\sigma' = \sigma - \frac{\text{tr}(\sigma)}{\text{tr}(\rho)} \rho$. Then $\text{tr}(\sigma') = 0$ and $(E - P_\rho)(\sigma') = \sigma'$. If $\sigma' \neq 0$, then 1 is an eigenvalue of $E - P_\rho$, contradicting the spectral gap $\rho_{\text{spec}}(E - P_\rho) < 1$. $\square$

Theorem 7.38 (Complement powers tend to zero). *If A is a primitive MPS tensor with PSD fixed point ρ , then $(\mathcal{E}_A - P_\rho)^n \rightarrow 0$ in operator norm.*

Proof. The spectral radius of $\mathcal{E}_A - P_\rho$ is strictly less than 1 by the spectral-gap hypothesis. Apply the general result that powers of an operator with spectral radius < 1 tend to zero. $\square$

Remark 7.39 (Definition mismatch). The Lean definition `IsPrimitiveMPS` encodes only the spectral-gap condition: there exists a nonzero PSD fixed point ρ and $\rho_{\text{spec}}(\mathcal{E}_A - P_\rho) < 1$. The paper’s notion of “primitive” ([SPGWC10, Proposition 3]) additionally requires ρ to be positive definite (full rank). Without this, `IsPrimitiveMPS` alone does *not* imply irreducibility or normality. A counterexample ($D = 2$, $\rho = \text{diag}(1, 0)$) is documented in the Lean source. To recover

the paper's full equivalence (primitive $\Leftrightarrow$ eventually full Kraus rank), one should assume ρ is positive definite or, equivalently, work with `IsIrreducibleTensor`. The remaining gaps are:

1. `IsPrimitiveMPS` + ρ positive definite $\Rightarrow$ `IsIrreducibleTensor` (requires formalising the restricted subchannel argument).
2. `IsIrreducibleTensor` $\Rightarrow$ `IsNormal` (requires Burnside's theorem for matrix algebras, not yet in Mathlib).

Chapter 8

Canonical Form Reduction

This chapter reduces a general normal MPS tensor to an injective block-diagonal form via iterated invariant-subspace decomposition. The reduction procedure follows [PGVWC07], [CPGSV21], and [Wol12, Ch. 6].

8.1 Block-diagonal assembly

Definition 8.1 (Block-diagonal tensor from blocks). Given r blocks with bond dimensions $(D_k)_{k=1}^r$, per-block MPS tensors $\{A_k^i\}$, and scaling factors $(\mu_k)_{k=1}^r$, define the block-diagonal tensor $A^i = \bigoplus_{k=1}^r \mu_k A_k^i$ of total bond dimension $D = \sum_k D_k$.

Theorem 8.2 (Per-block single-block FT). *If all block tensors A_k are injective and the per-block same-MPV condition holds (A_k and B_k generate the same MPV family for each k), then each pair (A_k, B_k) is gauge equivalent.*

Proof. Apply the single-block fundamental theorem (Theorem 3.13) to each block independently. $\square$

Theorem 8.3 (Per-block SameMPV implies global SameMPV). *If A_k and B_k generate the same MPV family for each block k , then the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ also generate the same MPV family.*

Proof. By the MPV decomposition (Theorem 2.22), each MPV is $\sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. Per-block equality of MPV coefficients gives global equality. $\square$

Theorem 8.4 (Global gauge from block gauge). *If each pair (A_k, B_k) is gauge equivalent ($B_k^i = X_k A_k^i X_k^{-1}$), then the block-diagonal tensors are gauge equivalent via the block-diagonal matrix $X = \bigoplus_k X_k$.*

Proof. Block-diagonal conjugation: $X(\bigoplus_k \mu_k A_k^i)X^{-1} = \bigoplus_k \mu_k X_k A_k^i X_k^{-1} = \bigoplus_k \mu_k B_k^i$. $\square$

8.2 Transfer map normalization

Theorem 8.5 (Scaling preserves injectivity). *If A is injective and $\zeta \neq 0$, then ζA is injective.*

Proof. Scaling by a nonzero scalar preserves linear independence and hence spanning. $\square$

Theorem 8.6 (Transfer map under scaling). $\mathcal{E}_{\zeta A}(X) = |\zeta|^2 \mathcal{E}_A(X)$.

Proof. Each summand: $(\zeta A^i)X(\zeta A^i)^\dagger = |\zeta|^2 A^i X (A^i)^\dagger$. $\square$

Theorem 8.7 (Phase-scaling preserves normalization). *If $\sum_i (A^i)^\dagger A^i = \mathbb{1}$ and $|\zeta| = 1$, then $\sum_i (\zeta A^i)^\dagger (\zeta A^i) = \mathbb{1}$.*

Proof. $(\zeta A^i)^\dagger (\zeta A^i) = \bar{\zeta} \zeta (A^i)^\dagger A^i = (A^i)^\dagger A^i$, since $\bar{\zeta} \zeta = |\zeta|^2 = 1$. $\square$

Theorem 8.8 (MPV under block normalization). *For any nonzero scaling factors μ_k , write $\mu_k = |\mu_k| \cdot (\mu_k/|\mu_k|)$. The MPVs of $\bigoplus_k \mu_k A_k$ and $\bigoplus_k |\mu_k| \cdot (\mu_k/|\mu_k|) A_k$ agree.*

Proof. By the MPV decomposition (Theorem 2.22), the k -th term is $\mu_k^N V^{(N)}(A_k)_\sigma$. The scaling identity (Lemma 2.12) absorbs the phase into the word evaluation. $\square$

8.3 Invariant subspace decomposition

Theorem 8.9 (Unitary conjugation preserves SameMPV). *If U is a unitary matrix, then A and $U^\dagger A U$ generate the same MPV family (as SameMPV₂).*

Proof. By cyclicity of the trace: $\text{tr}((U^\dagger A^{i_1} U) \dots (U^\dagger A^{i_N} U)) = \text{tr}(A^{i_1} \dots A^{i_N})$. $\square$

Definition 8.10 (Support projection). For a positive semi-definite matrix $\rho \geq 0$, the *support projection* P is the orthogonal projection onto the range of ρ . It is constructed via the spectral decomposition: if $\rho = U \text{diag}(\lambda_1, \dots, \lambda_D) U^\dagger$, then $P = U \text{diag}(\mathbf{1}_{\lambda_1 > 0}, \dots, \mathbf{1}_{\lambda_D > 0}) U^\dagger$, where $\mathbf{1}_{\lambda_j > 0}$ is 1 if $\lambda_j > 0$ and 0 otherwise.

Definition 8.11 (Invariant projection predicate). An MPS tensor A *has an invariant projection* if there exists an orthogonal projection P with $P \neq 0$, $P \neq \mathbb{1}$, and $(\mathbb{1} - P)A^i P = 0$ for all i . The negation of this predicate is IsIrreducibleTensor (Definition 8.14).

Theorem 8.12 (PSD fixed point gives invariant projection). *Let $\rho \geq 0$ be a fixed point of $\mathcal{E}_A$, and let P be the support projection of ρ . Then for all i , $(\mathbb{1} - P)A^i P = 0$: the range of P is invariant under all A^i .*

Proof. From $\mathcal{E}_A(\rho) = \rho$ and positivity of each summand $A^i \rho (A^i)^\dagger$, $(\mathbb{1} - P)A^i \rho (A^i)^\dagger (\mathbb{1} - P) = 0$ for each i . Since ρ is supported on the range of P , this forces $(\mathbb{1} - P)A^i P = 0$.

This is the finite-dimensional version of [Wol12, Prop. 6.10] (“Stationary subspaces”: the support projection of any stationary state is sub-harmonic, i.e. invariant under the channel). The equivalence between irreducibility and absence of nontrivial invariant projections is [Wol12, Thm. 6.2]; see also [CPGSV21, §IV.A.1]. $\square$

Theorem 8.13 (Two-block decomposition). *If the transfer map has a PSD fixed point ρ that is not positive definite (i.e. ρ has a nontrivial kernel), then A is unitarily equivalent to a two-block upper-triangular tensor. Moreover, the diagonal blocks generate the same MPV as A .*

Proof. The support projection P of ρ is neither 0 nor $\mathbb{1}$ (since $\rho \neq 0$ and ρ is not PD). Theorem 8.12 gives $(\mathbb{1} - P)A^i P = 0$ for all i . A unitary change of basis that block-diagonalizes P puts A^i in upper-triangular block form (Theorem 8.9 ensures the MPV is preserved by the conjugation). The trace of the upper-triangular product equals the sum of the block-diagonal traces, so the MPV decomposes into block contributions. $\square$

8.4 Iterated reduction

Definition 8.14 (Irreducible tensor). An MPS tensor A is *irreducible* (in the sense of [CPGSV21]) if the transfer map $\mathcal{E}_A$ has no nontrivial invariant projection, i.e. $\mathcal{E}_A$ is irreducible (Definition 4.22).

Theorem 8.15 (Irreducible block decomposition). *Every MPS tensor A can be decomposed (up to unitary equivalence) into a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ where each A_k is irreducible. The block-diagonal tensor generates the same MPV as A .*

Proof. By strong induction on the bond dimension D . If A is already irreducible, take $r = 1$. Otherwise, the transfer map has a PSD fixed point that is not PD (by the Cesàro argument), and Theorem 8.13 splits A into two blocks of strictly smaller bond dimension. Apply the induction hypothesis to each block and concatenate.

Both [PGVWC07] (Theorem 4, “TI canonical form”) and [CPGSV21] (§IV.A.1) obtain the block decomposition by the same invariant-projection splitting used here. An alternative route, developed in [Wol12, Thm. 6.14], characterises the fixed-point set of a quantum channel as a $*$ -algebra and then applies the Wedderburn–Artin decomposition ([Wol12, Eq. (1.39)]) to read off the block structure directly. Our formalization avoids this algebraic detour: we iterate the invariant-projection splitting with a termination proof by strong induction on D . $\square$

Theorem 8.16 (Diagonal PD fixed point). *If A is irreducible and trace-preserving, then the (unique) PSD fixed point of $\mathcal{E}_A$ is positive definite. After a unitary gauge transformation, the fixed point becomes a positive diagonal matrix.*

Proof. The irreducible tensor condition implies that $\mathcal{E}_A$ is irreducible as a CP map. By Theorem 5.2, the PSD fixed point is positive definite. The spectral theorem diagonalizes the PD matrix by a unitary conjugation. $\square$

8.5 Proportional single-block theorem

Theorem 8.17 (Proportional MPV implies gauge-phase equivalence). *Let A, B be injective normalized MPS tensors. If A and B generate proportional MPV families $(V^{(N)}(A))_\sigma = c_N V^{(N)}(B)_\sigma$ for some $c_N \in \mathbb{C}$ and $O_{AA}(N) \rightarrow 1$, $O_{BB}(N) \rightarrow 1$, then A and B are gauge-phase equivalent.*

Proof. By contradiction. If A and B are not gauge-phase equivalent, then $O_{AB}(N) \rightarrow 0$ (Theorem 6.17). From the proportionality hypothesis, $O_{AB}(N) = c_N O_{BB}(N)$. Since $O_{BB}(N) \rightarrow 1$, it follows that $c_N \rightarrow 0$. But $O_{AA}(N) = |c_N|^2 O_{BB}(N) \rightarrow 0$, contradicting $O_{AA}(N) \rightarrow 1$.

In [PGVWC07], the proportional case is handled as part of the canonical form argument. Our proof isolates it as a self-contained consequence of the spectral gap (Chapter 6). $\square$

8.6 Canonical form from primitivity

The canonical form predicate (Definition 9.8) includes an explicit hypothesis that the self-overlap $O_{A_k A_k}(N) \rightarrow 1$ for each block. The following theorem derives this from per-block primitivity, removing it as an independent assumption.

Theorem 8.18 (Canonical form from primitivity). *Let $(\mu_k, A_k)_{k=1}^r$ be a collection of block tensors and scaling factors satisfying:*

1. *each A_k is injective,*

2. each A_k is in DS gauge,
3. $|\mu_1| > \dots > |\mu_r|$ (strictly decreasing),
4. all $\mu_k \neq 0$,
5. each A_k is primitive (the transfer map $\mathcal{E}_{A_k}$ has peripheral spectrum $\{1\}$).

Then (μ_k, A_k) satisfies the canonical form predicate (Definition 9.8).

Proof. Conditions (1)–(4) are passed through directly. For condition (5) of the canonical form predicate ($O_{A_k A_k}(N) \rightarrow 1$), primitivity of $\mathcal{E}_{A_k}$ implies $\mathcal{E}_{A_k}^N(\rho) \rightarrow \rho_\infty$ for the unique fixed-point state ρ_∞ , and hence the self-overlap converges to 1. The Lean proof invokes `IsPrimitive.overlap_tendsto_one` for each block. $\square$

Remark 8.19. Theorem 8.18 means that the overlap convergence hypothesis in the canonical form predicate is *redundant* when primitivity is available. In the paper [CPGSV21], primitivity is obtained from irreducibility plus periodicity removal (blocking to period one). Theorem 8.20 formalizes this periodicity-removal step: under irreducibility and doubly-stochastic normalization, one can block until the transfer map becomes primitive. This corresponds to [CPGSV16, Appendix A].

Theorem 8.20 (Blocking yields a primitive transfer map). *Let A be an MPS tensor whose transfer map $\mathcal{E}_A$ is irreducible. Assume that A is doubly stochastic: $\sum_i A^i (A^i)^\dagger = \mathbb{1}$ and $\sum_i (A^i)^\dagger A^i = \mathbb{1}$. Then there exists a blocking length $p > 0$ such that the transfer map $\mathcal{E}_{A^{[p]}}$ of the blocked tensor $A^{[p]}$ is primitive.*

Proof. By Theorem 4.43, every peripheral eigenvalue of $\mathcal{E}_A$ is a root of unity. Since $\text{peripheral}(\mathcal{E}_A)$ is finite (Theorem 4.36), Lemma 4.40 gives a common exponent p such that $\mu^p = 1$ for all $\mu \in \text{peripheral}(\mathcal{E}_A)$. Lemma 4.41 then implies $\text{peripheral}(\mathcal{E}_A^p) = \{1\}$. Finally, the transfer map of the blocked tensor is the p -th power of the original transfer map, so $\mathcal{E}_{A^{[p]}}$ is primitive. $\square$

Remark 8.21. Theorem 8.20 is the “periodicity removal by blocking” step in [CPGSV16, Appendix A].

8.7 Canonical form existence pipeline ([CPGSV16])

The module `MPS/CanonicalFormExistence1606.lean` assembles the sorry-free components of the canonical-form construction following [CPGSV16, §2.3 and Appendix A].

Theorem 8.22 (Irreducible block decomposition (pipeline)). *Every MPS tensor A is SameMPV₂-equivalent to a block-diagonal tensor $\bigoplus_{k=1}^r A_k$ with each block irreducible. This re-exports Theorem 8.15.*

Proof. Direct application of Theorem 8.15. $\square$

Theorem 8.23 (CFII data for irreducible TP blocks). *Let A be an irreducible MPS tensor in the TP gauge ($\sum_i (A^i)^\dagger A^i = \mathbb{1}$) with $D > 0$. Then there exist a unitary U and a diagonal positive definite matrix Λ such that:*

1. $B^i := U^\dagger A^i U$ is TP,
2. $\mathcal{E}_B(\Lambda) = \Lambda$,

3. Λ is diagonal and positive definite,
4. A and B generate the same MPV family.

This is the “Canonical Form II” (CFII) step of [CPGSV16, Appendix A].

Proof. Apply Theorem 8.16 to obtain the unitary and the diagonal PD fixed point. Unitary conjugation preserves trace-preservation and the MPV family (Theorem 8.9). $\square$

Theorem 8.24 (Pipeline handoff: block decomposition with conditional CFII). *For any MPS tensor A , there exist irreducible blocks $(A_k)_{k=1}^r$ that are SameMPV₂-equivalent to A . Moreover, for each block k , assuming TP gauge and $D_k > 0$, one obtains CFII data (unitary, diagonal PD fixed point, MPV preservation).*

Proof. Combine Theorems 8.22 and 8.23. $\square$

Remark 8.25 (Missing steps for full canonical form existence). The pipeline in this section stops at the “handoff” point: irreducible blocks with *conditional* CFII data. The remaining steps for a complete canonical form existence theorem are:

1. *TP-gauge existence*: put each irreducible block into the TP gauge (Perron–Frobenius / spectral rescaling). This is not yet formalized.
2. *CFII $\rightarrow$ doubly-stochastic glue*: the CFII diagonal PD fixed point provides a weight for the adjoint fixed-point peripheral closure (Lemma 4.44), but the current blocking-primitivity theorem (Theorem 8.20) assumes the stronger doubly-stochastic condition. A generalization using the adjoint-fixed-point version would close this gap.
3. *Primitivity $\rightarrow$ normality/injectivity*: primitive blocks must be shown to be normal (and hence block-injective after further blocking). This requires either Burnside’s theorem or the Choi-representation spanning argument (see Remark 7.39 in Chapter 7).

Chapter 9

Block Permutation and Separation

This chapter establishes the algebraic structure of automorphisms of products of matrix algebras and uses it to prove that the same-MPV condition on block-diagonal tensors descends to per-block gauge equivalence. The approach follows [PGVWC07] and [CPGSV21].

9.1 Block permutation algebra

Theorem 9.1 (Ring isomorphisms permute block ideals). *Let $\{R_k\}_{k=1}^r$ be simple rings. Every ring isomorphism $T : \prod_k R_k \xrightarrow{\sim} \prod_k R_k$ permutes the block ideals: there exists a permutation σ such that T maps the k -th block ideal to the $\sigma(k)$ -th block ideal.*

Proof. Each block ideal $I_k = \{x : x_j = 0 \text{ for } j \neq k\}$ is an atom in the lattice of two-sided ideals (since R_k is simple). A ring isomorphism preserves the ideal lattice and hence permutes the atoms. Injectivity of the induced map on atoms gives the permutation σ . $\square$

Theorem 9.2 (Dimension preservation). *If T is an algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$, and σ is the induced permutation, then $D_{\sigma(k)} = D_k$ for all k .*

Proof. The restriction of T to the k -th block ideal induces a $\mathbb{C}$ -algebra isomorphism $M_{D_k}(\mathbb{C}) \cong M_{D_{\sigma(k)}}(\mathbb{C})$, which forces $D_k = D_{\sigma(k)}$ (by comparing dimensions as $\mathbb{C}$ -vector spaces). $\square$

Theorem 9.3 (Decomposition of automorphisms of $\prod M_{D_k}(\mathbb{C})$). *Every $\mathbb{C}$ -algebra automorphism of $\prod_{k=1}^r M_{D_k}(\mathbb{C})$ decomposes as a block permutation σ followed by per-block inner automorphisms: there exist $\sigma \in S_r$ with $D_{\sigma(k)} = D_k$ and invertible matrices $X_k \in \text{GL}_{D_k}(\mathbb{C})$ such that $T(M)_k = X_k M_{\sigma(k)} X_k^{-1}$.*

Proof. Theorem 9.1 gives the permutation σ . Theorem 9.2 gives $D_{\sigma(k)} = D_k$. The restriction of T to each block is a $\mathbb{C}$ -algebra automorphism of $M_{D_k}(\mathbb{C})$, which is inner by Skolem–Noether (Theorem 3.11).

This is a standard result in the theory of semisimple algebras; see [CPGSV21, §IV.A] for the MPS context. $\square$

9.2 Pi-algebra linear extension

Definition 9.4 (Per-block linear extension). Given injective tensors A_k, B_k of bond dimension D_k with the same MPV for each block k , the *per-block linear extension* T_k is the unique linear map $M_{D_k}(\mathbb{C}) \rightarrow M_{D_k}(\mathbb{C})$ satisfying $T_k(A_k^i) = B_k^i$ for all i . It exists (and is unique) because $\{A_k^i\}$ spans $M_{D_k}(\mathbb{C})$ by injectivity.

Definition 9.5 (Product algebra automorphism). The per-block linear extensions T_k are promoted to $\mathbb{C}$ -algebra homomorphisms (Lemma 3.12) and assembled into a single $\mathbb{C}$ -algebra automorphism $T : \prod_k M_{D_k}(\mathbb{C}) \xrightarrow{\sim} \prod_k M_{D_k}(\mathbb{C})$.

Theorem 9.6 (Per-block linear extensions assemble). *Let A_k and B_k be injective tensors of bond dimension D_k for $k = 1, \dots, r$, and suppose the per-block same-MPV condition holds. Then the per-block linear extensions $T_k : A_k^i \mapsto B_k^i$ assemble into an algebra automorphism of $\prod_k M_{D_k}(\mathbb{C})$, which decomposes as a permutation σ with $D_{\sigma(k)} = D_k$ and per-block conjugation matrices X_k .*

Proof. Each T_k exists uniquely and is multiplicative by Theorems 3.8 and 3.9. Promotion to algebra homomorphisms (Lemma 3.12) and assembly into the product gives an algebra automorphism. Theorem 9.3 decomposes it into permutation + per-block inner automorphisms. $\square$

Theorem 9.7 (Per-block SameMPV gives global gauge equivalence). *If all block tensors A_k are injective and per-block same-MPV holds, then each pair (A_k, B_k) is gauge equivalent, and the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ are globally gauge equivalent.*

Proof. Per-block gauge equivalence follows from the single-block FT (Theorem 3.13) applied to each block. Global gauge equivalence follows by assembling the per-block gauge matrices into a block-diagonal matrix (Theorem 8.4). $\square$

9.3 Block separation

Definition 9.8 (Canonical form predicate). A collection of scaling factors $(\mu_k)_{k=1}^r$ and block tensors $(A_k)_{k=1}^r$ satisfies the *canonical form predicate* if:

1. each A_k is injective,
2. each A_k is in DS gauge ($\sum_i (A_k^i)^\dagger A_k^i = \mathbb{1}$),
3. the moduli are strictly decreasing: $|\mu_1| > |\mu_2| > \dots > |\mu_r|$,
4. all $\mu_k \neq 0$,
5. the self-overlap converges: $O_{A_k A_k}(N) \rightarrow 1$ for each k .

Condition (5) is a primitivity hypothesis: for a primitive channel ([Wol12, Thm. 6.7, item 3]), $T^n(\rho) \rightarrow \rho_\infty$ for every initial state, which forces the self-overlap to converge to 1. See also [Wol12, Thm. 6.8] for the completely-positive characterisation and Chapter 6 for the spectral-gap proof used in this formalization.

Remark 9.9. In the literature [CPGSV21], condition (5) is derived from primitivity of the transfer map (peripheral spectrum = $\{1\}$). Theorem 8.18 now formalizes this derivation: per-block primitivity implies overlap convergence, so condition (5) is redundant when primitivity holds. The canonical form predicate retains the explicit hypothesis for flexibility (it can be satisfied without proving primitivity of each block).

Theorem 9.10 (Vandermonde separation). *If $\mu_1, \dots, \mu_r$ are pairwise distinct complex numbers and $\sum_{k=1}^r c_k \mu_k^N = 0$ for $N = 0, 1, \dots, r-1$, then $c_k = 0$ for all k .*

Proof. The coefficient vector c satisfies $Vc = 0$ where V is the Vandermonde matrix of the μ_k . Injectivity of μ makes V invertible. $\square$

Lemma 9.11 (Block separation core). *Under the hypotheses of the canonical form predicate (strict ordering, DS gauge, self-overlap convergence), if $\sum_k \mu_k^N (V^{(N)}(A_k)_\sigma - V^{(N)}(B_k)_\sigma) = 0$ for all N, σ , then A_k and B_k generate the same MPV for each k .*

Proof. By induction on the number of blocks r . In the inductive step, take overlaps with the leading block A_0 ; cross-block contributions from $k \geq 1$ decay exponentially (by the spectral gap), leaving $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \rightarrow 0$. The proportional FT (Theorem 8.17) gives $A_0 \simeq B_0$. Peeling off the leading block and dividing by μ_0^N , the induction hypothesis applies. $\square$

Theorem 9.12 (Block separation from canonical form). *Let $((\mu_k), (A_k))$ satisfy the canonical form predicate, and let (B_k) be injective tensors in DS gauge. If the block-diagonal tensors $\bigoplus_k \mu_k A_k$ and $\bigoplus_k \mu_k B_k$ generate the same MPV family, then each pair (A_k, B_k) generates the same MPV family.*

Proof. By induction on the number of blocks r .

Base case ($r = 1$): the summed identity $\mu_1^N (V^{(N)}(A_1)_\sigma - V^{(N)}(B_1)_\sigma) = 0$ and $\mu_1 \neq 0$ give A_1 and B_1 same MPV.

Inductive step ($r \geq 2$): the leading block 0 has the largest $|\mu_0|$. Taking overlaps with a test tensor A_0 and using the spectral gap (Theorems 6.17 and 6.15), the cross-overlap contributions from non-leading blocks decay to zero. This shows $O_{A_0 A_0}(N) - O_{A_0 B_0}(N) \rightarrow 0$, and since $O_{A_0 A_0}(N) \rightarrow 1$ (by the primitivity hypothesis), the proportional FT (Theorem 8.17) gives A_0 and B_0 gauge-phase equivalent, hence same MPV.

Peeling off the leading block and dividing by μ_0^N , the induction hypothesis applies to the remaining $r - 1$ blocks.

In [PGVWC07], the uniqueness of the canonical form is proved differently: the argument uses OBC (open boundary condition) canonical forms together with a Horn–Johnson matrix-equation lemma ([HJ91, Thm. 4.4.14]). In [CPGSV21], the block separation is deduced from the BNT uniqueness (Theorem 10.5). Our formalization makes the separation explicit by induction on the number of blocks r , with each step using the spectral gap to peel off the dominant block. $\square$

Theorem 9.13 (Canonical form fundamental theorem). *Under the hypotheses of Theorem 9.12, there exist a permutation σ and per-block gauge matrices X_k such that $B_k^i = X_k A_{\sigma(k)}^i X_k^{-1}$.*

Proof. Theorem 9.12 gives per-block same-MPV. Theorem 9.7 converts this to per-block gauge equivalence with permutation. $\square$

Chapter 10

BNT and Full Assembly

This chapter introduces *Basis Normal Tensors* (BNT), establishes their permutation rigidity, and assembles the full Fundamental Theorem of Matrix Product States. The results combine the canonical form reduction (Chapter 8), block separation (Chapter 9), spectral gap (Chapter 6), and Newton–Girard identities into the final statement. The main references are [PGVWC07] and [CPGSV21].

10.1 Basis normal tensors

Definition 10.1 (Basis Normal Tensor (BNT)). A *basis normal tensor* (BNT) decomposition for a total tensor A_{tot} consists of g block tensors $(A_1, \dots, A_g)$ with bond dimensions $(D_1, \dots, D_g)$ such that:

1. each A_j is normal (Definition 2.18),
2. at each system size N , the MPV of A_{tot} is a linear combination of the block MPVs:
 $|V^{(N)}(A_{\text{tot}})\rangle = \sum_j c_j |V^{(N)}(A_j)\rangle$,
3. for sufficiently large N , the block MPV vectors $\{|V^{(N)}(A_j)\rangle\}_{j=1}^g$ are linearly independent.

This is [CPGSV21, Definition 4.2].

Definition 10.2 (Canonical form with BNT separation). An *IsCanonicalFormBNT* extends the canonical form predicate (Definition 9.8) with the additional requirement that distinct blocks are *not* gauge-phase equivalent: for $j \neq k$ with $D_j = D_k$, A_j and A_k are not gauge-phase equivalent.

Theorem 10.3 (Cross-overlap decay for distinct CF-BNT blocks). *If $((\mu_k), (A_k))$ satisfies the CF-BNT predicate and $j \neq k$, then $O_{A_j A_k}(N) \rightarrow 0$.*

Proof. If $D_j \neq D_k$ the overlap is zero by dimension mismatch. If $D_j = D_k$, the blocks are not gauge-phase equivalent (by the CF-BNT separation condition), so the spectral-gap theorem (Theorem 6.17) gives decay to 0. $\square$

Theorem 10.4 (CF-BNT yields BNT). *If $((\mu_k), (A_k))$ satisfies the CF-BNT predicate, then the block tensors form a BNT decomposition for the block-diagonal tensor $\bigoplus_k \mu_k A_k$.*

Proof. **Normality:** each block is injective (hence 1-block injective, hence normal). **Spanning:** the MPV decomposition (Theorem 2.22) gives $V^{(N)}(A_{\text{tot}})_\sigma = \sum_k \mu_k^N V^{(N)}(A_k)_\sigma$. **Linear independence:** the overlap matrix $G_{jk}(N) = O_{A_j A_k}(N)$ converges to the identity (diagonal entries $\rightarrow 1$ by the self-overlap hypothesis; off-diagonal entries $\rightarrow 0$ by the spectral gap, since distinct CF-BNT blocks are not gauge-phase equivalent). A Gram matrix converging to the identity is eventually invertible, giving linear independence. $\square$

10.2 BNT permutation rigidity

Theorem 10.5 (BNT permutation — proportional MPV case). *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two BNT decompositions such that the block-diagonal tensors generate proportional MPV families with convergent nonzero coefficients. Then $g_A = g_B$, and there is a permutation $\sigma \in S_g$ with $D_{\sigma(k)} = D_k$ and $A_{\sigma(k)}$ gauge-phase equivalent to B_k for each k .*

Proof. From the proportional MPV hypothesis and convergent coefficients, one extracts (for each pair (j, k)) the limit $\lim_{N \rightarrow \infty} O_{A_j B_k}(N)$. By the spectral gap (Chapter 6), cross-block overlaps between non-equivalent blocks decay to 0. The resulting overlap matrix has at most one nonzero entry per row and per column, which defines a permutation. Nonzero overlap forces gauge-phase equivalence.

This is part of [CPGSV21, Theorem 4.1] (the Fundamental Theorem for proportional MPVs), which establishes the permutation and gauge-phase equivalence from the uniqueness of BNT decompositions (characterised in [CPGSV21, Proposition 4.1]). $\square$

Theorem 10.6 (BNT permutation — overlap-orthonormal case). *Let $(A_j)_{j=1}^{g_A}$ and $(B_k)_{k=1}^{g_B}$ be two families of injective DS-gauge tensors whose self-overlaps converge to 1 and whose cross-overlaps (within each family) converge to 0. If they span the same MPV subspace at every system size, then $g_A = g_B$ and the blocks are related by a permutation and gauge-phase equivalence.*

Proof. The overlap matrix $O_{A_j B_k}(N)$ converges to a matrix with at most one nonzero entry per row/column (by the spectral gap). This is the “primitive branch” of the argument, where the overlap orthonormality hypotheses are assumed directly rather than derived from proportional MPVs with convergent coefficients. $\square$

10.3 Coefficient convergence

Theorem 10.7 (Coefficient ratio decay). *Under the canonical form predicate, for $k \geq 1$, $(\mu_k/\mu_0)^N \rightarrow 0$ as $N \rightarrow \infty$.*

Proof. Since $|\mu_k| < |\mu_0|$ (strict ordering of moduli), $|\mu_k/\mu_0| < 1$, so the geometric sequence converges to 0. $\square$

10.4 Newton–Girard identities

Theorem 10.8 (Newton–Girard trace recursion). *If $A, B \in M_n(\mathbb{C})$ satisfy $\text{tr}(A^k) = \text{tr}(B^k)$ for all $k \geq 1$, then A and B have the same characteristic polynomial.*

Proof. The Newton–Girard identities express the coefficients of the characteristic polynomial recursively in terms of the power-sum traces $\text{tr}(A^k)$. Equal power sums give equal coefficients, hence equal characteristic polynomials. $\square$

Theorem 10.9 (Equal power sums imply equal multisets). *Let $\alpha, \beta : \{1, \dots, n\} \rightarrow \mathbb{C}$. If $\sum_i \alpha_i^k = \sum_i \beta_i^k$ for all $k \geq 1$, then the multisets $\{\alpha_i\}$ and $\{\beta_i\}$ are equal.*

Proof. Construct diagonal matrices $\text{diag}(\alpha)$ and $\text{diag}(\beta)$. The power-sum hypothesis gives $\text{tr}(\text{diag}(\alpha)^k) = \text{tr}(\text{diag}(\beta)^k)$. By Theorem 10.8, the characteristic polynomials agree: $\prod_i (X - \alpha_i) = \prod_i (X - \beta_i)$. Extracting roots gives multiset equality. $\square$

10.5 Primitivity bridge

Definition 10.10 (Primitive MPS tensor). A *primitive MPS tensor* is one equipped with a nonzero PSD fixed point ρ of the transfer map such that the complementary spectral radius $\rho(\mathcal{E}_A - P) < 1$ (where P is the fixed-point projection).

Theorem 10.11 (Primitive overlap convergence). *If A is a primitive MPS tensor, then $O_{AA}(N) \rightarrow 1$.*

Proof. Immediate from the primitive overlap convergence (Theorem 6.22). $\square$

10.6 Full assembly

Theorem 10.12 (Fundamental Theorem — proportional MPV case). *Let $((\mu_k^A), (A_k))$ and $((\mu_k^B), (B_k))$ be two CF-BNT decompositions. If the block-diagonal tensors generate proportional MPV families, then:*

1. $r_A = r_B$ (same number of blocks),
2. there exists a permutation σ with $D_{\sigma(k)}^A = D_k^B$,
3. each pair $(A_{\sigma(k)}, B_k)$ is gauge-phase equivalent.

Proof. The CF-BNT predicate produces a BNT decomposition (Theorem 10.4). Coefficient convergence (Theorem 10.7) provides the convergent nonzero coefficients. The BNT permutation theorem (Theorem 10.5) gives the block count, permutation, and per-block gauge-phase equivalence. $\square$

Theorem 10.13 (Fundamental Theorem — equal MPV case). *Under the same hypotheses as Theorem 10.12, if the MPV families are equal (not just proportional), then the block-diagonal tensors are gauge equivalent (not just gauge-phase equivalent).*

Proof. Theorem 10.12 gives the permutation and per-block gauge-phase equivalence. From equal MPVs, the power-sum identity $\sum_k (\mu_k^A)^N = \sum_k (\mu_k^B)^N$ holds for all N . By Theorem 10.9, the multisets $\{\mu_k^A\}$ and $\{\mu_k^B\}$ agree. Combining with the permutation and per-block gauge-phase equivalence gives global gauge equivalence.

This is the full Fundamental Theorem of MPS as stated in [CPGSV21, Theorem 4.1] and originally proved in [PGVWC07]. The key deviation from the literature is the explicit use of Newton–Girard identities (Theorem 10.8) to pass from equal power sums of scaling factors to multiset equality. $\square$

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